Building and Evaluating a Cross Hazard Machine Learning Approach to Unified Lunar Forecasting

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Abstract

LIPAS (Lunar Impact and Predictive Analysis System) is a hybrid forecasting and decision support platform that combines machine learning and physics-based computational platform designed to predict the Moon's space weather conditions. These conditions include temperature, radiation, dust activity, micrometeorite impacts, solar storms, and moonquakes. By unifying space-weather data products from NASA and the National Oceanic and Atmospheric Adminstration (NOAA) alongside real-time Operating Missions as Node on the Internet (OMNI) telemetry and historical archives, the LIPAS platform processes data through an temporal Long Short Term Memory (LSTM) neural network. Users can view the LIPAS space-weather predictions via an interactive, web-based dashboard for any lunar location.

Firstly, it explains step by step how the innovative hybrid model (LIPAS) was built and what was learned during the building process. Secondly, it studies how operational usability can be improved in real mission scenarios compared to existing prediction systems. Thirdly, this research investigates whether integrating machine learning and probabilistic modeling into a single platform (LIPAS) improves predictions accuracy compared to existing single-purpose (NASA & NOAA) lunar environmental models.

The ultimate goal of this research project is to highlight how combining scientific research, advanced software engineering, and diverse datasets creates a practical tool (e.g LIPAS) that supports safer and more predictable lunar exploration in the future.

1. Introduction

Renewed international interest in lunar exploration, particularly through NASA's Artemis program and its international partners, has increased the need for reliable and accessible information about the lunar environment. The Moon has no protective atmosphere or global magnetic field, leaving its surface exposed to large temperature variations, radiation, micrometeorite impacts, electrostatic dust, and seismic activity. These hazards can affect both crew

safety and the operation of spacecraft and surface equipment.

A range of scientific models already exists for studying individual lunar hazards. However, these models are generally designed for specific hazards and operate as separate tools. This makes it difficult to combine their outputs into a single operational view or use them together for forecasting. Mission planners may therefore need to consult several different data sources and models when assessing environmental conditions, which can make the decision-making process more difficult.

LIPAS (Lunar Impact and Predictive Analysis System) was developed to address this fragmentation by bringing multiple lunar hazard forecasts into a single decision-support platform. The system combines machine-learning predictions with physics-based models, public heliophysics data, and adaptive weighting between the two approaches. Based on the literatute review, LIPAS is among the first publicly documented lunar forecasting frameworks to combine replicated agency physics, live heliophysics feeds, adaptive hybrid weighting, and machine-learning inference within a unified operational decision-support architecture.

This study addresses three research questions. First, can a hybrid lunar forecasting system be developed by combining machine learning and physics-based approaches effectively? Second, can integrating multiple lunar environmental hazards and data sources into a unified forecasting and decision-support system improve operational usability? Third, can the final LIPAS hybrid system improve prediction performance for selected lunar environmental hazards compared with established physics-based and replicated NASA and NOAA reference models?

The aim of LIPAS is not to replace established scientific models, but to combine their outputs within a common forecasting framework. The main contribution of this work is therefore the development and evaluation of a unified hybrid system that retains physics-based constraints while using machine learning where it can provide additional predictive information. In addition to forecasting multiple hazards within one platform, the system is intended to make these forecasts easier to interpret and use in mission-planning workflows.

2. Overview of Current Models

2.1 General Overview

The study of lunar environmental conditions has historically relied on single-purpose models, each focused on a particular hazard. While these models are developed with advanced scientific understanding in their respective domains, they remain limited when applied to operational mission planning, as they neither provide integrated insights nor real-time predictive support.

Each model was built to solve a specific scientific problem, not to work together as part of a broader forecasting system. This

creates a gap as the models are unable to analyze the interactions of these hazards.

2.2 NASA DONKI Model

The NASA Database of Notifications, Knowledge, Information (DONKI) API provides near real-time monitoring of solar flares, coronal mass ejections, and solar energetic particle events (NASA, 2023). DONKI is valuable as a space-weather event catalog for radiation context but is only event-driven and primarily reactive, offering limited temporal coverage and incomplete data for continuous mission planning. It does not produce multi-day forecasts or integrate with other hazard types, making it insufficient as a standalone planning tool.

2.3 Temperature Models

Temperature predictions rely heavily on the Diviner Lunar Radiometer Experiment (Williams et al., 2017; NASA, 2023), which delivers detailed infrared measurements of surface temperatures across the Moon. While accurate for thermal modeling on the Moon where the terrain is highly uneven, they do not themselves produce short-term forecasts. Similarly, Langseth et al. (1972) provide Apollo-era heat-flow measurements, yet these datasets are sparse and lack temporal resolution for short-term forecasts. Neither source was designed to produce future temperature predictions, only to describe past and present conditions.

The Diviner model calculates localized dayside surface temperatures based on solar insolation physics. The baseline dayside formula is defined as:

$$T = 392 \cdot \cos(z)^{0.25} - 273$$

where **T** represents the surface temperature in degrees Celsius, and **z** represents the solar zenith angle relative to the target coordinates. For nightside temperature profiles, legacy models rely on a polynomial fit derived from Apollo-era heat-flow measurements. While mathematically deterministic, these standalone formulas do not dynamically adjust.

2.4 Lunar Dust Models (LADEE LDEX)

The lunar dust environment has been studied through the Lunar Atmosphere and Dust Environment Explorers Lunar Dust Experiment mission (LADEE LDEX) (Horányi et al., 2015), which documents dust particle density and electrostatic activity. While critical for understanding surface hazards, LDEX data is limited to the LADEE mission epoch and orbital coverage, offering neither continuous global monitoring nor predictive capabilities. The LADEE mission ended in 2014, so no new measurements have been produced since, leaving significant gaps in temporal and spatial coverage.

2.5 Moonquakes Apollo PSE

Moonquake monitoring relies on the Apollo Passive Seismic Experiment (Onodera et al., 2024), which identified thousands of previously undetected moonquakes. However, the PSE dataset is retrospective, unable to provide real-time forecasting, and does not integrate with other hazards. Despite this limitation, the catalog contains rich statistical patterns tied to the lunar tidal cycle, which LIPAS uses as physics/catalog context for moonquake rate estimates. Because deep moonquakes occur in clusters correlated with the Earth-Moon tidal stress cycle, these repeating patterns can be identified in the historical catalog and projected forward in time, giving LIPAS a statistical basis for forecasting even without real-time seismic sensors.

2.6 Micrometeorite Prediction MEM

Micrometeorite impact fluxes are addressed by the NASA Meteoroid Engineering Model (MEM 3) (Moorhead et al., 2020), which provides estimates of collision rates. MEM is precise within its narrow domain but does not incorporate secondary effects, such as dust lofting or interactions with solar activity, which may influence surface operations. LIPAS therefore adds shower-calendar modulation and dust coupling on top of the MEM 3 baseline used in its physics path.

2.7 GOES-18 Radiation

The NOAA Geostationary Operational Environment -18 satellite provides continuous monitoring of solar X-ray flux and energetic particle counts (NOAA, 2023). The GOES-18 satellite was developed for a highly accurate Earth orbit radiation forecast system, but data must first be adapted for lunar surface conditions, as the Moon's lack of a magnetosphere means that radiation exposure is significantly different from what is measured in near-Earth orbit. GOES-18 provides no multi-day forecast capability and was not designed with lunar missions in mind.

2.8 Conclusion

Although individual models such as DONKI, Diviner, MEM, PSE, and LDEX are essential for scientific research, their limitations in coverage, integration, and predictive usability highlight the need for a unified system.

LIPAS was developed to address these gaps by combining the strengths of existing single-purpose models into a multi-hazard, unified forecasting system. By integrating real-time NASA and NOAA space weather data, historical lunar measurements, and developing a new kind of hybrid machine learning model to intelligently use that data, LIPAS enhances operational usability and situational awareness. Unlike previous models, LIPAS is both proactive and integrative: delivering a fast and comprehensive tool for mission planning and safety assessment.

3. System Overview (LIPAS)

LIPAS is a hybrid forecasting system that combines machine learning with physics-based calculations. Rather than using machine learning for every hazard, LIPAS dynamically degociates trust between neural predictions and physics based equations in real time. This allows established physical relationships to remain the main basis for hazards that are already well described by physics, while machine learning can provide additional corrections where relationships between changing environmental variables may contain useful predictive information.

This new LIPAS architecture provides a scalable, less risk prone system that may be vital for future crewed and uncrewed missions by space agencies such as NASA and ESA.

3.1 Prototype Limitations and the Move to a Backend

The original version of LIPAS was a browser-based application in which data processing and machine learning were performed on the client side. This worked for the initial multi-hazard prototype, but problems became apparent as the system became more complex, because it relied solely on machine learning, disregarding the strong physical relationships between hazards. Temperature was one example: different training runs produced variable forecasts, while a simpler physics-based calculation remained more stable and accurate overall. This showed that applying machine learning to every hazard was not necessarily beneficial.

To fix these flaws, the app was completely re-engineered into a server-assisted hybrid model, with a Python backend. The heavy data filtering, API polling, and model training are now completely handled on the server, which serves as a stable data engine that feeds a polished, fast-rendering front-end dashboard. LIPAS is now able to decide whether to use machine learning or physics based on what is more accurate.

3.2 Hybrid Forecasting and Uncertainty Architecture

At the heart of the new backend is the core forecasting script, space_weather_model.py. The model uses an extended 18-feature input structure. These features describe the environmental conditions used by the forecasting system and include hazard-related, temporal, spatial, and space-weather variables. The model does not use only the conditions at one moment. Instead, it processes a 72-step temporal window, meaning that it receives a sequence of previous observations and model inputs when producing a prediction. This allows the neural network to use information about how conditions have changed over time.

The model produces six target outputs: solar activity, radiation, temperature, moonquake rate, micrometeorite flux, and dust density. Although there are six outputs, the model uses 18 inputs, allowing information from one environmental variable to contribute to the prediction of another hazard.

But as previously explained LIPAS does not only rely on the neural network. A physics_estimate function first calculates a

physics-based estimate for each relevant hazard using the physical relationships described in the Methods section. This provides a physically grounded starting point for the prediction. A refine_prediction function then combines this estimate with the neural-network output, allowing machine learning to make a correction rather than having to learn the entire physical relationship from the data.

For each hazard channel i, the final hybrid prediction $\hat{Y}_{hybrid,i}$ is calculated as a weighted sum of the machine learning output and the physics baseline:

$$\hat{Y}_{hybrid,i} = w_i \hat{Y}_{ML,i} + (1 - w_i)\hat{Y}_{Physics,i}$$

Hybrid calculation

The weight is calculated during prediction using the current space-weather conditions and an estimate of the neural network's uncertainty.

$$w_i = \max\left(0,\ \min\left(1,\ (W_{base,i} + \Delta_{regime,i}) \cdot \max\left(0.35,\ 1 - 1.6\frac{\sigma_{ML,i}}{S_i}\right)\right)\right)$$

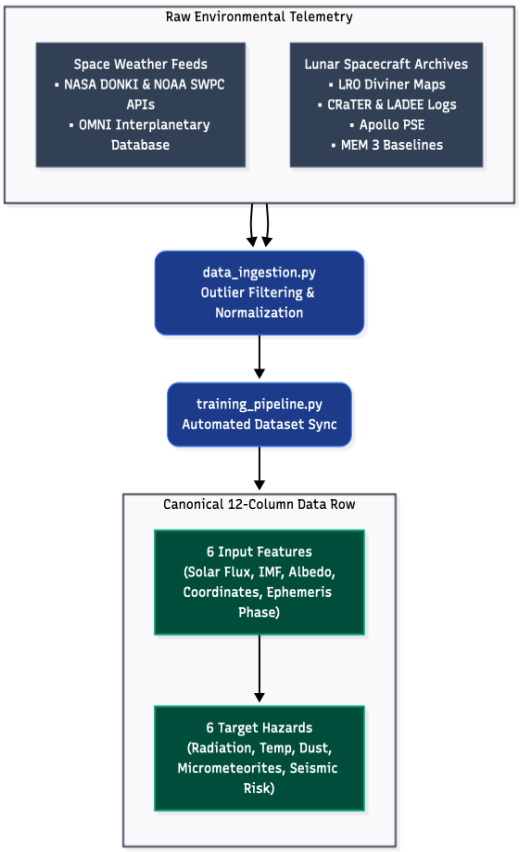
Hybrid mix

In this formulation, $W_{base,i}$ represents the baseline machine learning weight for a specific hazard, and $\Delta_{regime,i}$ applies an algorithmic shift based on the current storm posture, such as increasing the model weight during volatile solar events where machine learning excels. The term $\Delta_{regime,i}$, represents the epistemic uncertainty meaning uncertainty caused by the model's limited knowledge of the underlying relationship, particularly when it encounters conditions that differ from those represented in its training data. This uncertainty is derived from the Monte Carlo dropout standard deviation (the model is run multiple times with some neural-network connections randomly switched off, and the variation between those predictions is used to estimate how uncertain the model is), which is normalized by S_i ,the physical operational span of the hazard channel. This mathematical architecture is the core of the platform's operational safety. If the neural network encounters out-of-distribution (OOD) data, meaning inputs that differ substantially from the conditions represented in its training data**,** and becomes highly uncertain, the $\sigma_{ML,i}$ value increases. This mathematically demotes the neural network's influence and forces the system to safely fall back on the established physics baselines.

Where established physics provides a strong basis for a hazard, LIPAS gives greater weight to the physics-based path rather than relying primarily on machine learning. In the locked production blend, temperature, moonquakes, micrometeorite flux, and dust are mostly physics-led, while solar activity and radiation retain a higher machine-learning weights.

To handle real-world data gaps and sensor volatility, models.py also supports optional ensemble forecasting for uncertainty estimation, a key feature for mission planners. The active production path, however, serves a single locked temporal model for uncertainty-focused runs.

The Two-Step Hybrid Core & Ensemble Refinement Architecture



3.3 Development of the LIPAS Hybrid Framework

The final version of LIPAS had been implemented using HTML, CSS, and JavaScript for the front end, with a dedicated Python TensorFlow backend responsible for data ingestion, model training, physics blending, and forecast serving.

Multiple production training runs were completed during development. The final quantitative studies used the second training run model because, at the time of evaluation, because, at the time of evaluation, it was the best-performing completed production model. Its chronological held-out test RMSE was approximately 4.97, compared with approximately 13.66 for the corresponding persistence baseline, and no later completed training run had beaten it.

Earlier prototype experiments used the browser-based neural network that preceded the Python hybrid backend. These experiments are treated as developmental diagnostics rather than evidence of the final production model's performance, as they lacked the architecture and baseline framework used in the final evaluation.

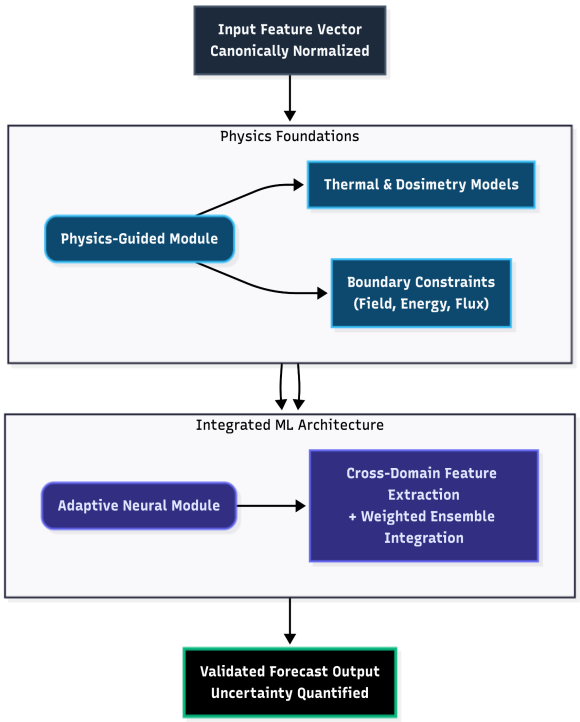
The final production model was then evaluated using the five-study framework described in Section 4.

3.4 Streamlined Data Ingestion and Pipeline Orchestration

The backend handles real-time data streams through a modular pipeline script called data_ingestion.py, which queries, cleans, and normalizes external space agency APIs into the system's extended 18-feature layout. The pipeline unifies completely different source families, processing real-time coronal mass ejection events from the NASA DONKI API, solar indices from the NOAA Space Weather Prediction Center, and interplanetary magnetic field data from the OMNI database. These live feeds are automatically These space-weather feeds are combined with lunar data sources, including LRO Diviner thermal maps, CRaTER

radiation logs, and LADEE/LDEX dust measurements. For long-term historical baselines, the script integrates Apollo passive seismic experiment data, Apollo surface heat flow measurements, the Meteoroid Engineering Model, and peer-reviewed micrometeorite flux datasets. The entire end-to-end data flow is managed by training_pipeline.py, which coordinates the automated collection schedule, runs the model training loops, and applies a strict two-sigma outlier detection filter which flags values that fall more than two standard deviations from the expected range of the data. These values are treated as potential anomalies and blocked from entering the model.

The Unified Data Ingestion and Canonical Formatting Pipeline



3.5 Command Center Dashboard and Map Visualization Engine

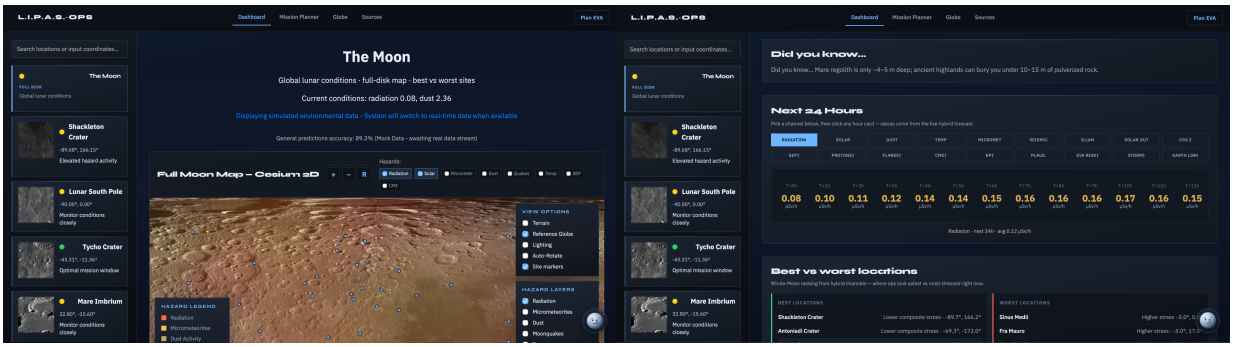
The user interface was completely overhauled after the limitations of the prototype model, were identified in the premirlary studies. This transformed the original basic webpage into a mission-control-style interface based on CSS Grid and Flexbox to ensure it would be easy to use. The LIPAS application consists of three main pages: the dashboard, mission planner, and Cesium globe.

The dashboard includes a sidebar containing more than 50 lunar locations, as well as a coordinate search bar that allows users to view detailed hazard metrics for any selected location. The dashboard provides topographic maps with hazard activity, detailed 30-day predictions, proton and particle flux visualizations, and clear warnings and recommendations to support mission planning and hazard awareness (see next page). Historical and predicted hazard trends are also visualized using Chart.js, allowing users to switch between line, bar, and area representations.

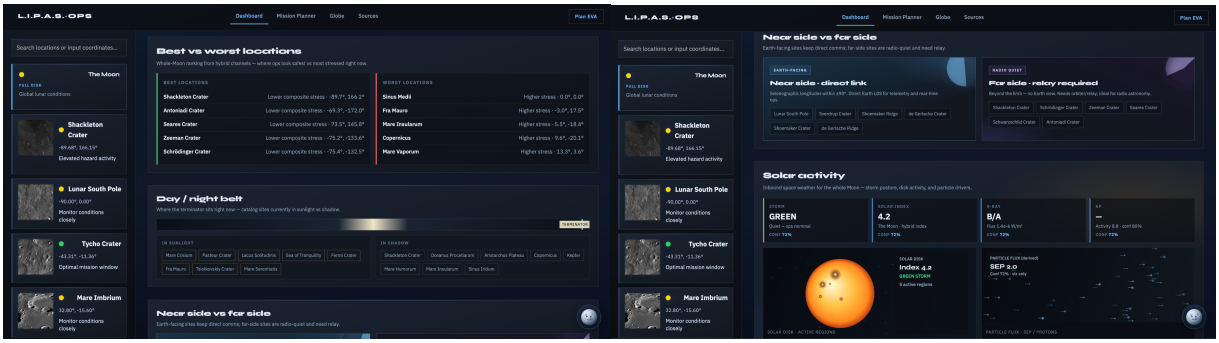
The mission-planning interface allows users to enter mission parameters and duration targets. Based on the available environmental forecasts, the system can generate suggested EVA departure windows, assess landing-zone conditions, and produce an exportable mission plan. The LIPAS Luna Assistant further supports mission planners by providing additional information based on the available hazard data.

The Cesium map page provides an interactive 3D or 2D lunar map with hazard overlays and corresponding timelines. This

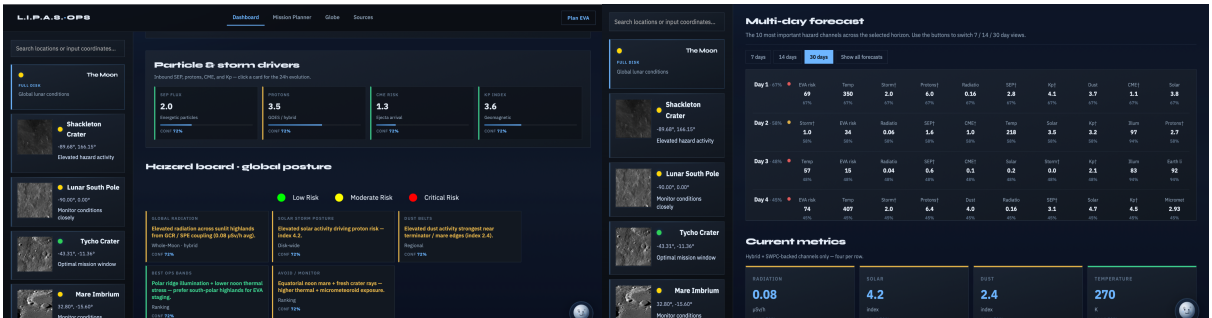
allows users to view the environmental conditions across the lunar surface and quickly identify how different hazards vary by location and over time.



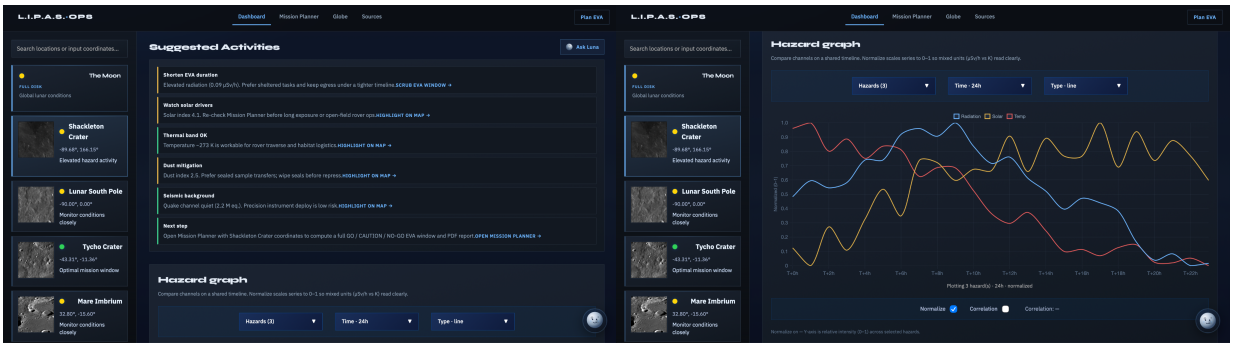
LIPAS operator dashboard for a selected lunar site, showing the location sidebar with coordinate search, a site-specific topographic hazard map with current and forecast overlays, and the 24-hour forecast panel (https://lipas.netlify.app).



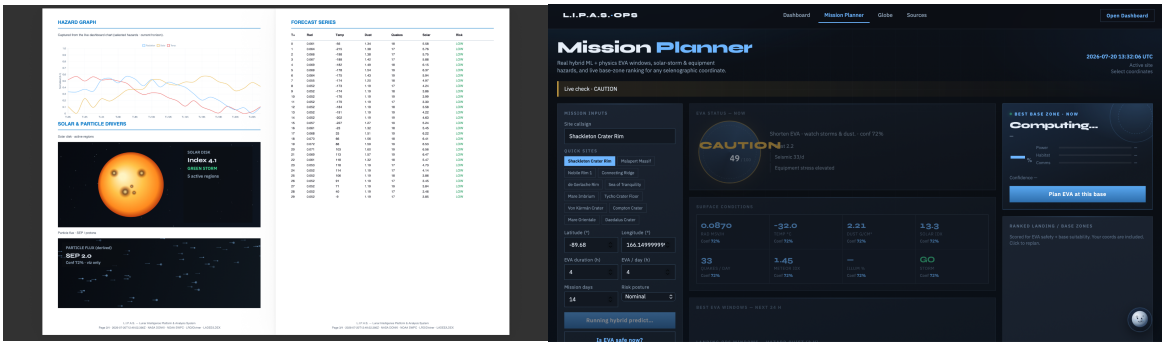
LIPAS situational panels for the selected site, including multi-location hazard comparison, the solar terminator line, communications strength, and animated solar-activity context.



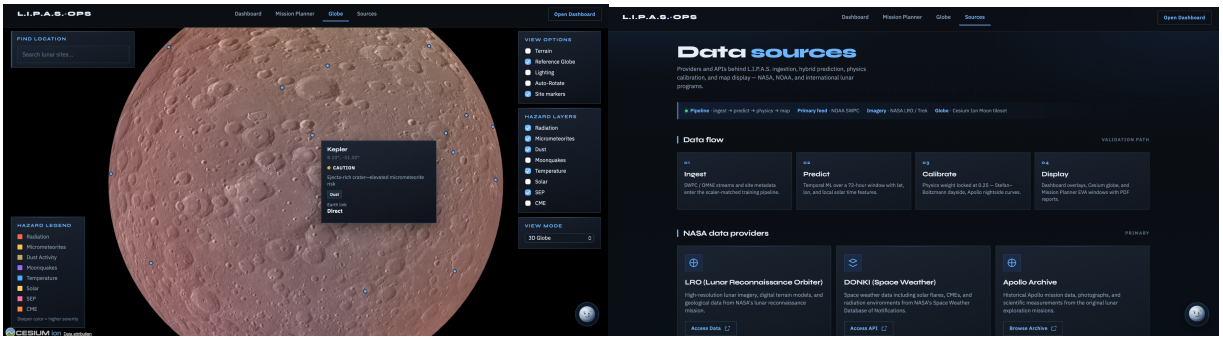
LIPAS forecast and diagnostics view, showing storm drivers, current hazard status, outlooks up to 30 days, and an extended panel of all 18 calculated metrics.



LIPAS decision-support panels showing ML-suggested activities and a normalized multi-hazard graph for comparing hazard interactions over time.



LIPAS mission-planning outputs, including a downloadable PDF report and an ML-assisted EVA scheduler that estimates operational risk, radiation exposure, and preferred activity windows.



Interactive Cesium lunar globe with site selection and time-varying hazard overlays, shown alongside the LIPAS sources page listing all 48 data sources used by the system.

4. Methodology and Experimental Design

4.1 Experimental Design and Evaluation Framework

The research was divided into three phases, with each phase addressing one of the three research questions. The first phase focused on the development of LIPAS and what was learned during its development. The second examined its operational use in a simulated mission environment. The third evaluated whether the final hybrid system could improve prediction performance compared with established physics-based and agency-style reference models. All final quantitative evaluations used the locked production model from the second training run.

Phase 1 – Development of the LIPAS Hybrid Model

The first phase examined how LIPAS was developed from its original prototype into the final hybrid system. This included the early fourteen-day prototype evaluation, which was conducted during daytime conditions at 0°N, 0°E from 5–19 April 2026. The purpose of this test was not to establish the final accuracy of LIPAS, but to identify weaknesses in the original client-side machine-learning approach and determine which parts of the system required a physics-based foundation.

The prototype results showed that treating all six hazards as machine-learning problems could produce unstable results for hazards that are strongly determined by physical relationships. These findings led to the development of the hybrid architecture used in the final studies. Temperature, moonquakes, dust, and micrometeorites were therefore kept mainly physics-based, while machine learning was given a larger role in solar activity and radiation.

The development phase also included the creation of the final data pipeline, the 18-feature input structure, the 72-step temporal input window, the physics-based prediction paths, and the hybrid weighting system. These components were then combined into the locked production model used in the remaining phases.

Phase 2 – Operational Mission Evaluation

The second phase investigated whether integrating several environmental hazards into one system could improve operational usability during a simulated lunar mission.

A controlled radiation-storm EVA simulation was created using a timeline based on the duration of Apollo 17 EVA 3, approximately seven hours and 37 minutes. During the simulation, the LIPAS forecast was monitored at fixed time intervals and compared with the information an operator would receive by consulting separate public NASA and NOAA products, including GOES, NOAA SWPC Space Weather Prediction Center, and NASA DONKI data.

The same predefined caution and no-go thresholds were applied to the different forecasting pathways. The comparison focused on when each system produced warnings and when the simulated EVA would reach a caution or no-go condition. The purpose was not to establish that LIPAS is safer than NASA systems, but to examine whether combining different sources of environmental information into one forecasting system can provide useful additional warning information and reduce

the need to interpret several separate products manually.

Phase 3 – Quantitative Prediction Evaluation

The third phase tested whether the final LIPAS hybrid model could improve prediction performance compared with established physics-based and replicated NASA and NOAA reference models.

Two final fourteen-day evaluation windows were used at three lunar locations: 0°N, 0°E, Apollo 11, and Nobile Rim. Window A covered 5–18 July 2026 and represented a lunar-night period, while Window B covered 22 July–4 August 2026 and represented a lunar-day period.

For each study date and location, LIPAS generated predictions for six hazards: solar activity, radiation, temperature, moonquake rate, micrometeorite flux, and dust density. The predictions were compared with persistence, physics-only references generated using the same physics_estimate path used during forecasting, and replicated public NASA and NOAA baselines described in Section 4.3.

The evaluation used archived OMNI, NOAA, GOES, and NASA DONKI data for the study dates. This allowed the experiments to be replayed using the same inputs rather than depending on changing live data. LIPAS and the reference models were therefore evaluated under the same dates, locations, input conditions, and evaluation targets.

Because continuous co-located measurements of all six hazards are not available across the lunar surface, each evaluation channel was classified as measured, estimated, or physics-based proxy, meaning an indirect estimate constructed from physical relationships when direct measurements were unavailable, before scoring. Prediction performance was primarily assessed using root mean square error (RMSE) and mean absolute error (MAE). RMSE gives greater weight to larger errors, while MAE represents the average absolute difference between predictions and evaluation targets.

The micrometeorite analysis included an additional comparison during Window B. LIPAS used the MEM 3 seasonal background together with AMS-based (American Meteor Society, that tracks meteoroids) meteor-shower modulation, while the replicated MEM 3 comparison did not include the additional shower modulation. This allowed the effect of explicitly representing meteor-shower timing to be examined during the Perseid period. The result was therefore interpreted as a test of shower modulation rather than as evidence of general superiority over the complete NASA micrometeorite modelling framework.

Finally, an ablation analysis was performed to examine how different inputs contributed to the hybrid predictions. Selected hazard and environmental inputs were removed while the remaining conditions were kept fixed. The resulting predictions were compared with the full hybrid model using the same RMSE and MAE measures. The

ablation conditions included the full hybrid model, removal of solar and storm drivers for the radiation evaluation, removal of micrometeorite context for the dust evaluation, an ML-only condition, and a physics-only condition. This analysis was used to determine how much the different environmental inputs contributed to the final predictions and whether information from one hazard could influence another.

Together, these three phases were designed to answer the three research questions from different perspectives: how LIPAS should be built, whether its integrated structure is useful for mission operations, and whether the resulting hybrid system provides measurable predictive improvements over established reference approaches.

4.2 Baseline Configurations and Control Environments

To ensure a completely fair and objective evaluation, specific baseline rules were established across all testing environments. The flagship references used for comparison were gathered from public databases, peer reviewed formulas, engineering model exports, and official agency documentation (see bibliography for more information). No internal NASA forecast product was assumed. If an established flagship model did not provide direct multi day lunar predictions for a specific timeframe, the research replicated the underlying public NASA or NOAA method as accurately as possible from published formulas and open data products, as detailed in Section 4.3. For every hazard, the study recorded the source, the procedure, and whether the comparison target was a direct measurement, an engineering baseline, or a physics proxy, reflecting the limited availability of continuous lunar measurements, which is one of the motivations for LIPAS. For the mission simulation and ablation tests, the control environment held coordinates and physical constants steady while selected inputs were altered. The mission simulation used real archived space weather context for the chosen EVA day and checked hazard status at fixed intervals along the scripted timeline. The ablation tests blocked specific driver or hazard inputs completely to observe system behavior. By isolating these controls, the experimental framework measured performance limits and cross hazard dependence.

4.3 Baseline Replication Procedures and Model Specifications

This subsection explains how NASA models (not necessarily the actual systems NASA uses to make predictions, but models compiled of open end data) were replicated as accurately as possible. For further information on different models please refer to section 2.

Temperature baselines were replicated using lunar thermal physics based on Diviner observations and Apollo nightside constraints. On the dayside, surface temperature was computed from a Stefan Boltzmann balance,

$$T = \left(\frac{(1-\alpha) I_0 \cos\theta}{\varepsilon\sigma} \right)^{1/4}$$

where **I** is the sunlight reaching the Moon, adjusted for the Earth–Sun distance on that day, **a** is the surface reflectivity (albedo), **ε** is how well the surface gives off heat (emissivity), **σ** is the Stefan–Boltzmann constant, and **μ** represents the Sun's angle above the surface based on the location and local time. Nightside temperatures were calculated using an Apollo-based cooling curve that estimates how the surface cools after sunset. The model also adjusts for latitude and includes a small amount of heat from the lunar interior. Surface properties such as reflectivity, emissivity, terrain, and thermal inertia were taken from LIPAS's site data. When Diviner temperature data were available, they could be added to improve the estimate. The early prototype also used the simpler Diviner-style dayside formula:

$$(1-\alpha) I_s = \varepsilon\sigma T^4$$

This was included for continuity with the early developmental test, while the final second-train studies used the Stefan–Boltzmann dayside model and Apollo-based nightside model as the main temperature baselines.

Radiation and solar activity baselines were built from public space-weather data rather than from a direct lunar radiation forecast. Solar activity used OMNI2 and SWPC data, including Kp, IMF Bz, solar wind speed and density, and F10.7 when available. NASA DONKI records for flares, CMEs, geomagnetic storms, and SEP events were also used for the study dates. Radiation used a CRaTER-based galactic cosmic-ray background of about 0.38 mSv per day, converted to an hourly value and adjusted for solar activity. Additional quiet, storm, and SEP terms were based on Kp, southward IMF Bz, high solar-wind speed, GOES proton data, and solar activity. A small latitude-based increase was also applied near the lunar poles. GOES X-ray and proton measurements were therefore used as inputs to the lunar radiation model, not as direct measurements of radiation on the Moon.

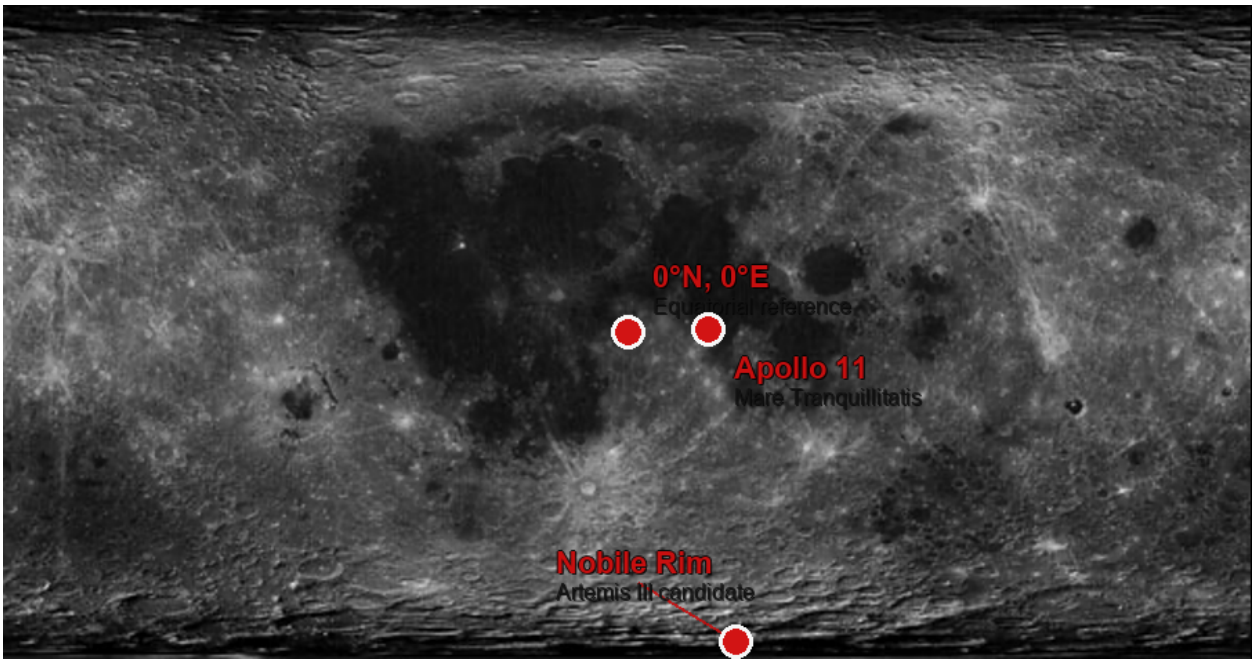
Micrometeorite baselines were replicated from NASA MEM 3 using the project's MEM 3 export. The annual MEM 3 flux provided the main background level, while the available high- and low-density ranges were used to model seasonal changes. LIPAS added AMS-based meteor-shower modulation separately from the basic MEM 3 baseline. Latitude-based adjustments were also applied only in the LIPAS physics path. For Window B, the flagship baseline used MEM 3 without the AMS shower term, while LIPAS used MEM 3 with the additional shower modulation. This allowed the effect of meteor-shower activity to be tested directly.

Dust baselines were based on LADEE LDEX data rather than a live global dust forecast. The model combined a normal background level with dust produced by micrometeorite impacts, increased dust near the lunar terminators, an optional LDEX local-time factor, and a small effect from regolith depth. This made dust a physics

proxy based on LDEX observations rather than a claim of continuous dust measurements at the Artemis sites.

Moonquake baselines were based on the Apollo Passive Seismic Experiment catalog. The model combined deep, shallow, and thermal moonquake rates, then added stronger activity near the terminators, a small equatorial preference for deep events, and a weak regolith effect. No real-time lunar seismometer network was assumed. The result was therefore treated as a catalog-based physics proxy rather than a direct measurement of current moonquake activity at the study sites.

Persistence baselines, which assume the latest known value stays constant, and physics-only references were generated from the same replay data. RMSE and MAE were calculated for LIPAS against persistence, the physics-only references, and each replicated flagship baseline. The mission simulation used the same LIPAS Model outputs and baseline systems throughout the Apollo 17-length EVA timeline. The ablation study used the same replay data while removing selected inputs or model pathways. Together, these procedures made the quantitative studies reproducible using public data, published formulas, and the project's MEM 3 export.



Moon Map showing the locations LIPAS studies were based on in red.

5. Results

5.1 Qualitative Comparison

The qualitative comparison uses a structured framework to classify the capabilities of each system across nine criteria.

Criteria Definitions:

1. Hazard Coverage: the range of distinct lunar environmental hazards the system addresses.

2. Data Fusion Capability: the degree to which the system integrates data from multiple sources.

3. Temporal Forecasting Ability: the capacity to produce predictions for future time periods.

4. Spatial Coverage: the geographic extent of the lunar surface the system can address.

5. Real-time Operational Capability: the ability to incorporate current data feeds and produce timely outputs.

6. Robustness and Stability: the consistency and reliability of system outputs under varying conditions.

7. Validation Strength: the degree to which system outputs have been compared against observed data.

8. Usability and Operational Readiness: the accessibility of the system for non-specialist mission planning use.

9. Openness and Accessibility: the degree to which the system and its data are publicly available.

Qualitative Scale:

Labels used: None, Very Low, Low, Moderate, High, Very High. These are categorical labels only and are not assigned numerical values.

System Classification:

Before the comparison, each system is classified by its primary function. This ensures that systems designed for different purposes are understood in context rather than compared as if they were built for the same goal.

DONKI: Event-based monitoring system.

Diviner: Measurement and archival dataset.

LDEX: Measurement system with limited geographic and temporal scope.

Apollo PSE: Historical measurement catalog.

MEM 3: Statistical engineering model.

GOES-18: Real-time measurement system adapted from Earth-orbit context.

LIPAS: Integrated predictive system combining measurement, modeling, and machine learning. (uses data from all the systems mentioned above)

Qualitative Comparison Table

Model	Hazard Coverage	Data Fusion	Temporal Forecasting	Spatial Coverage	Real-time Ops	Robustness	Validation	Usability	Openness
LIPAS	Very High	Very High	High	High	High	Moderate	Moderate	Very High	High
DONKI	Low	Low	Low	Moderate	High	High	High	Moderate	High

Diviner	Very Low	Low	Low	Very High	Low	Very High	Very High	Low	Moderate
LDEX	Very Low	Low	None	Low	None	Moderate	High	Very Low	Moderate
Apollo PSE	Very Low	Low	None	Very Low	None	High	Very High	Low	High
MEM 3	Very Low	Low	Moderate	High	Low	High	High	Low	Moderate
GOES-18	Low	Moderate	Moderate	Moderate	Very High	Very High	Very High	Moderate	High

The qualitative matrix reveals a clear trade-off between specialized scientific detail and multi-hazard operational utility. Traditional flagship models and data archives, such as the Diviner Lunar Radiometer and the Apollo Passive Seismic Experiment (PSE), exhibit the highest possible marks in *Robustness and Stability* and *Validation Strength* This is expected, as they represent foundational, peer-reviewed space agency instrumentation with long operational histories. However, these traditional tools are limited by their design by *None* or *Very Low* scores in *Hazard Coverage*, *Data Fusion*, and *Temporal Forecasting*, requiring mission planners to look at multiple separate data packets.

In contrast, LIPAS addresses these structural gaps by achieving *Very High* ratings in *Hazard Coverage*, *Data Fusion Capability*, and *Usability and Operational Readiness*. By integrating six different environmental variables into a single browser-based user interface, it provides an immediate operational overview that individual models miss. The primary trade-off of this integrated approach is reflected in LIPAS 's Moderate scores for *Robustness* and *Validation Strength*. Because LIPAS relies heavily on live public API feeds and has currently only been validated against three 14-day localized datasets, its validation and reliability is lower than that of historical, non-predictive science archives. This demonstrates that while LIPAS does not replace the scientific depth of flagship physics models, it provides a broader real-time predictive decision-support capability for mission planning.

Limitations of this qualitative framework: all classifications are based on available documentation and published descriptions of each system. Some criteria involve judgment at boundaries, particularly between Moderate and High. The framework reflects the systems as they are currently used in operational or research contexts, not their maximum theoretical capability.

5.2 Preliminary Accuracy Studies and Model Redesign

This study evaluates the premirlary quantitative forecasting performance of LIPAS V1 over two independent fourteen-day windows using replicated NASA and NOAA baseline models. The methodology and baseline construction are described in Section 4.3.

The first was a fourteen day multi hazard study from April 5 to April 19 under daytime conditions at coordinates 0°N 0°E, tracking all six lunar hazards against NASA and NOAA flagship models. The second was a short term versus long term temperature study during that same window, evaluating the platform against the NASA Diviner dayside calculations. The main goal of these

older tests was to find the strengths and weaknesses of the integrated system before making final adjustments to the platform code. These early studies were intentionally limited: they left out many secondary environmental factors, used only a subset of NASA's publicly available models and datasets, and did not fully replicate the additional mathematical work already developed by physicists across the separate hazard models, because their purpose was diagnostic, not final validation.

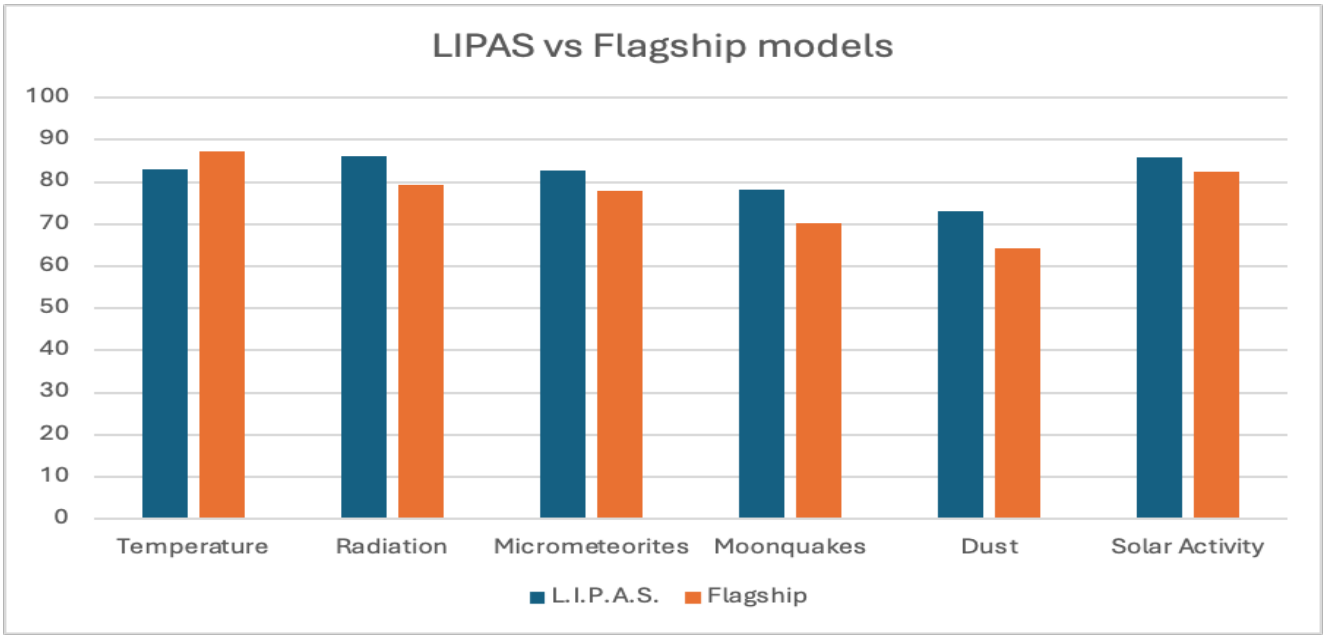
The results from the initial multi hazard study showed that combining multiple data streams gave the early system a major advantage in tracking complex environment patterns. Compared to standalone flagship models, the early platform achieved RMSE values approcimately 10.3 times better at predicting radiation, 4.8 times better for solar activity, 2.5 times better for moonquakes, 1.9 times better for micrometeorites, and 1.8 times better for dust. These ratios apply only to that early prototype comparison and should not be treated as final skill scores for the current hybrid backend. This confirmed that the cross hazard design could successfully use context from one hazard to improve the tracking of another, providing more consistent forecasts.

However, this old study also revealed a severe flaw in the thermal component, where temperature predictions performed 1.8 times worse than the flagship models. The neural network used a machine learning model that was far too advanced for a simple variable like temperature. When paired with the physical formulas, the neural network overfitted the historical data and created less accurate results than a clean mathematical equation.

The secondary short term versus long term temperature study confirmed this issue. For this baseline comparison, the real Diviner dayside formula was used. While the system showed smaller variation in its outputs compared to the standalone Diviner model, temperature remained the only area where the integrated system underperformed against standalone space agency methods. Predicting further into the future showed that the overcomplicated machine learning layer could not outperform a clean, direct mathematical formula. Full data tables for this old short term versus long term temperature study are located in Appendix A.

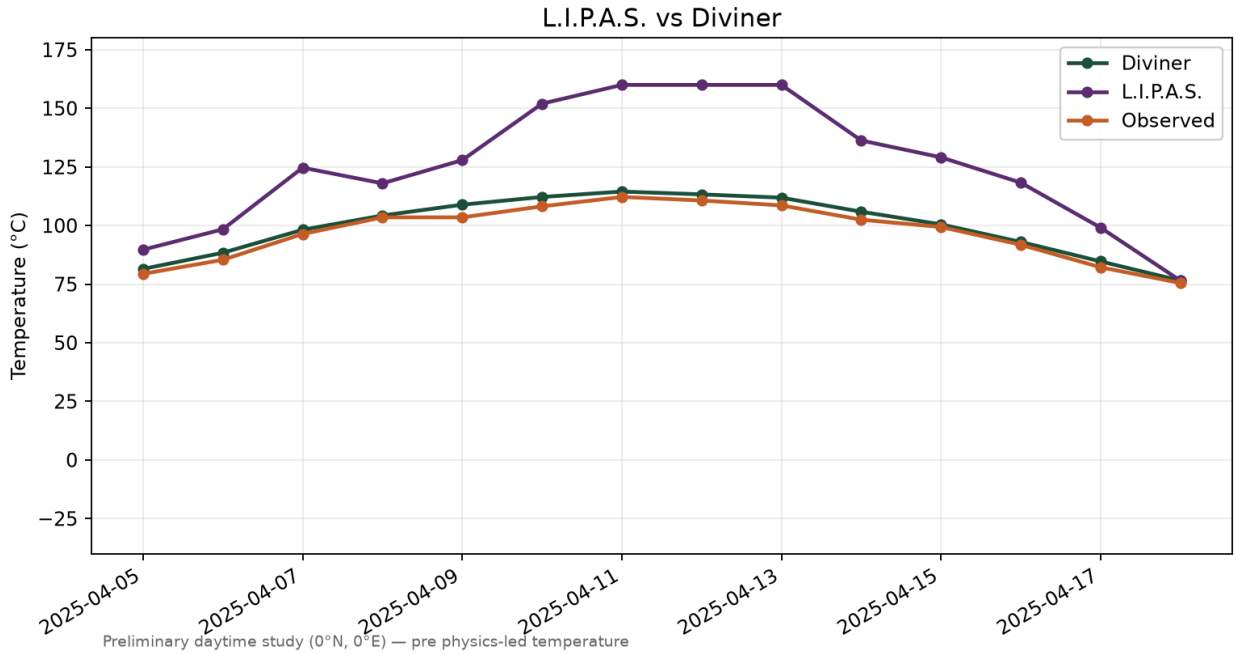
Seeing the data from these premirlary studies was highly valuable because it revealed exactly where the model was failing, which prompted a complete overhaul of the platform code. To fix the temperature issues and prevent overfitting, the architecture of the system was drastically modified to better balance the machine learning layers with raw physical equations. The full technical details and developments of these substantial improvements are documented in the system overview section. These old findings also showed that using loose linear approximations for baseline models was insufficient, leading to a complete redesign of the final experimental setup to use published space agency formulas, denser multi-source inputs, and a fairer hybrid comparison framework for all subsequent evaluations.

Figure 1 summarizes the early multi-hazard comparison under daytime conditions at 0°N, 0°E.



Preliminary LIPAS versus flagship scores across six hazards (5–19 April, daytime, 0°N 0°E). Early prototype results only; not the final Window A evaluation.

Figure 2 shows the same window's temperature track against Diviner and observed labels.



Preliminary daytime temperature at 0°N, 0°E. Early client-side LIPAS diverges from Diviner and observed labels; the thermal overfit case that later motivated physics-led temperature.

5.3 Final Multi-Hazard Evaluation and Performance

This study evaluates the real quantitative forecasting performance of LIPAS V2 over two independent fourteen-day windows using replicated NASA and NOAA baseline models. The methodology and baseline construction are described in Section 4.3.

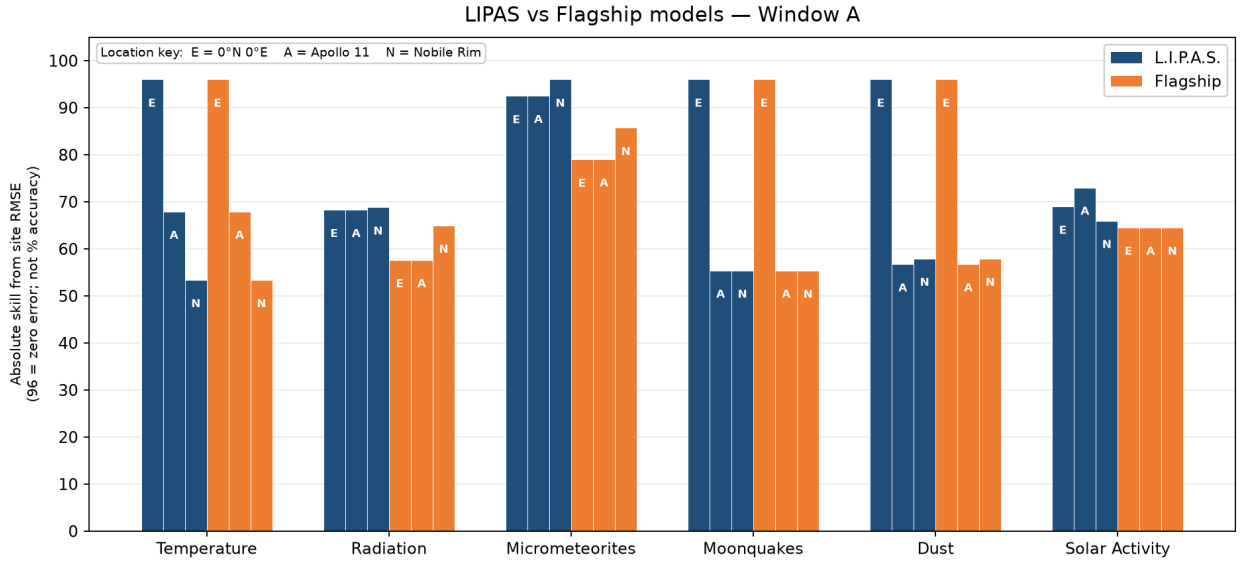
The LIPAS study was from 5 July through 18 July 2026 and 22 July to 4 August. The same three sites were used for both planned windows A and B: an equatorial reference at 0°N, 0°E (comparison for prototype), Apollo 11 in Mare Tranquillitatis, and the Artemis III Nobile Rim candidate. Together they give forty-two site-days.

For window A (first study) local solar time at the study longitudes fell mostly on the nightside during this fortnight (about 19 percent dayside samples), so Window A is treated as the night regime.

Against the flagship replicas, the hybrid improved solar-activity RMSE (3.95 versus 4.66, about 15 percent) and radiation RMSE (0.102 versus 0.132, about 22 percent). Temperature, moonquakes, and dust remain hybrid–flagship ties because both use the same physics path, while absolute site RMSE still differs (equatorial near-perfect where labels match physics; Apollo and Nobile retain larger residuals). Micrometeorites are not a tie: with AMS coupling versus a no-shower MEM 3

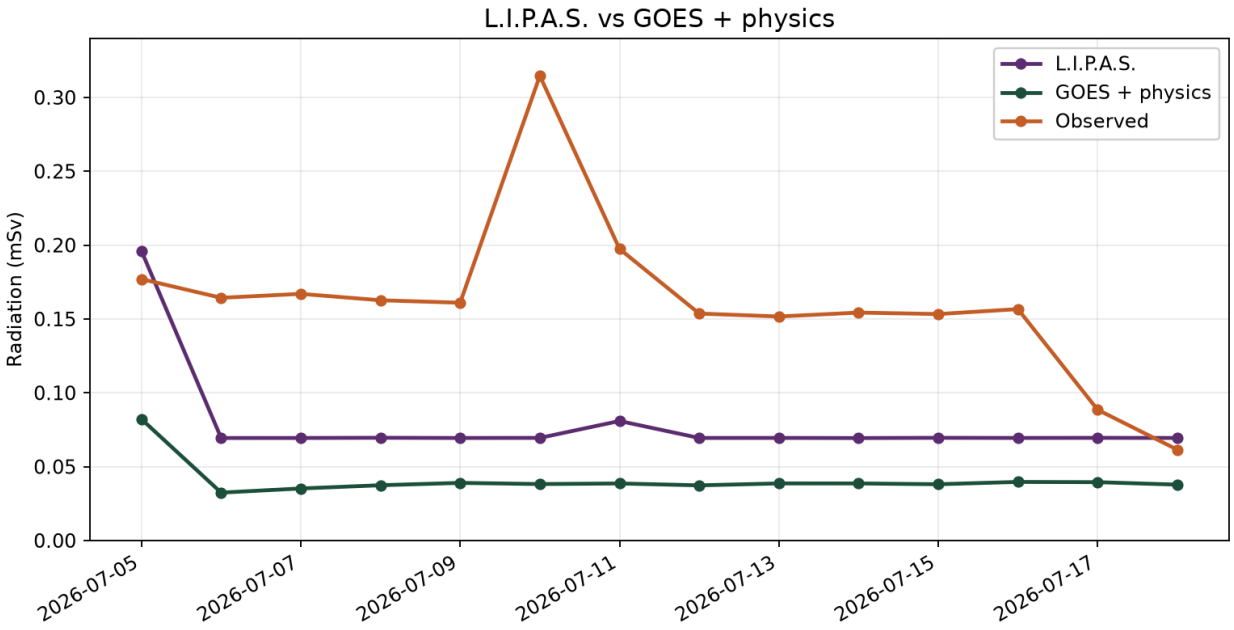
flagship, RMSE is 0.011 versus 0.109 under mild July shower timing. Radiation skill is nearly site-invariant because space-weather drivers dominate; LIPAS solar skill varies modestly by site, while the flagship solar replica is identical at all three sites by construction.

Figure 3 shows Window A absolute skill from per-site RMSE (not percent accuracy). LIPAS improves solar activity and radiation. Temperature, moonquakes, and dust show clear E/A/N structure even though LIPAS and Flagship match at each site. Micrometeorites favor LIPAS at every site, with a small Nobile advantage from the latitude term. Flagship solar bars are flat because that public replica is heliospheric (meaning estimates are driven by the Sun's global environment rather than local conditions on the Moon).



Window A (5–18 July 2026; 42 site-days). Absolute skill from per-site RMSE (96 ≈ zero error; not percent accuracy). E = 0°N 0°E; A = Apollo 11; N = Nobile Rim. Blue = L.I.P.A.S.; orange = Flagship.

Figure 4 shows the radiation time series at the equatorial reference site over the same night window. The hybrid track stays closer to the evaluation labels than the flagship replica across most days.



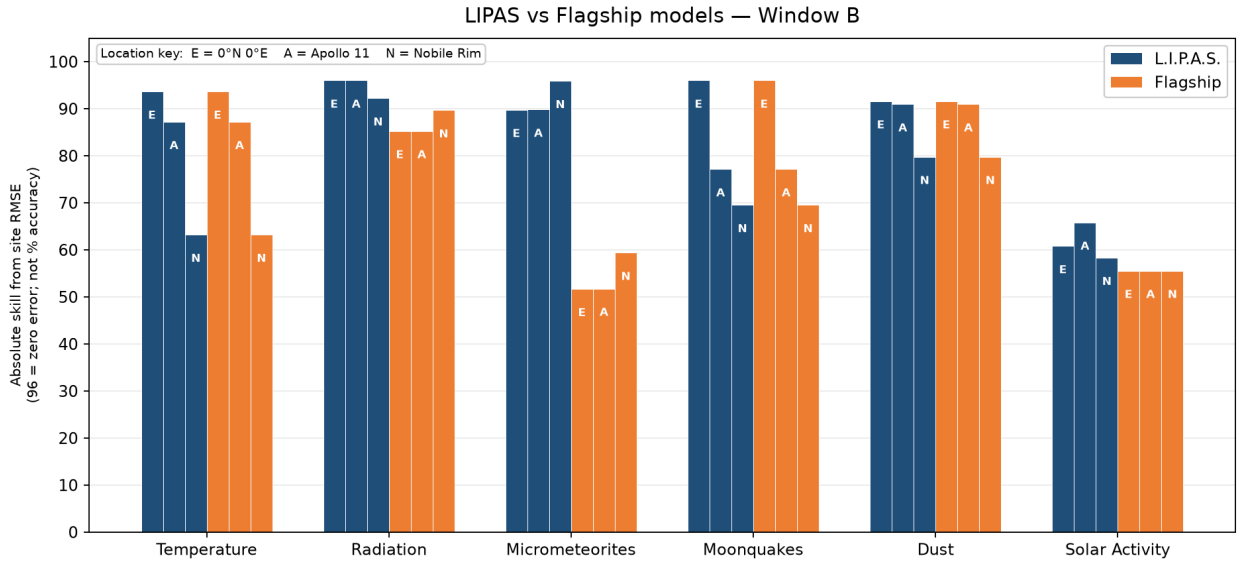
Window A radiation at Apollo 11 landing site (5–18 July 2026). LIPAS hybrid versus flagship replica (and truth labels). This channel is highlighted because it showed the clearest hybrid accuracy gain under nightside conditions.

Hazard	Hybrid	Flagship	Relative change	Outcome
Solar activity	3.95	4.66	−15%	Hybrid better
Radiation (mSv)	0.102	0.132	−22%	Hybrid better
Temperature (°C)	38.2	38.2	0%	Tie (shared physics; site RMSE varies)
Moonquakes (day^{-1})	1.23	1.23	0%	Tie (shared physics; site RMSE varies)
Micrometeorite flux ($\times 10^{-15}$)	0.011	0.109	−90%*	Hybrid better (AMS coupling 0.90)
Dust (g cm^{-3})	0.526	0.526	0%	Tie (shared physics; site RMSE varies)

Table 1. Window A hybrid versus flagship RMSE (5 July - 18 July 2026); lunar-night dominant; 42 site-days).

Table 1 reports Window A hybrid versus flagship RMSE. Temperature, moonquakes, and dust are model ties on the shared physics path. Micrometeorites favor LIPAS under AMS coupling (0.011 versus 0.109). Solar activity and radiation improve by about 15 and 22 percent.

Window B used the same sites, locked model, blend, and metrics on the complementary lunar-day fortnight: 22 July through 4 August 2026. Local solar time was dayside on about 95 percent of site-days, so Window B is the day regime. For micrometeorites in this window, LIPAS used the MEM 3 seasonal envelope plus AMS-style shower modulation, including Perseids (peak day-of-year 216, 4 August). The flagship used the MEM 3 seasonal envelope without those shower bumps.



Window B (22 July–4 August 2026; 42 site-days). Absolute skill from per-site RMSE (96 ≈ zero error; not percent accuracy). Same site letters as Window A. Flagship solar is the samea across sites (heliospheric). Micrometeorites separate because LIPAS includes AMS showers while the flagship does not.

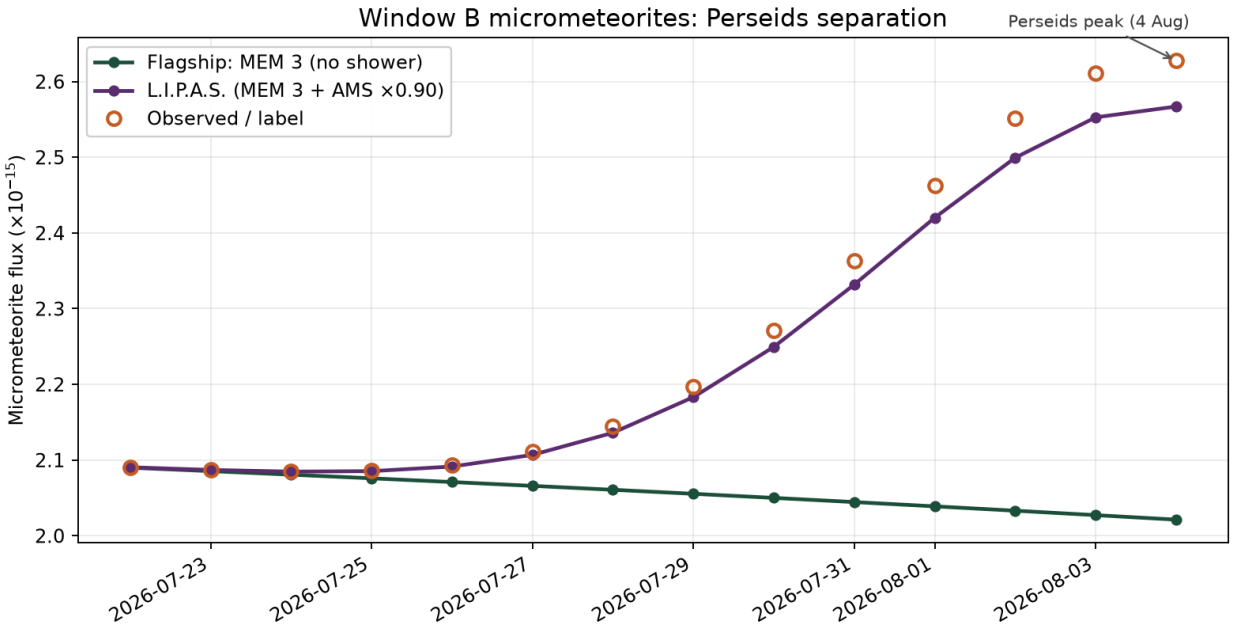
Under those dayside conditions, LIPAS again improved solar activity (5.09 versus 5.99 RMSE, about 15 percent). Temperature, moonquakes, and dust remain ties on the shared physics path, with site skill still varying under the Window B labels. Micrometeorites are not a new scoring rule in Window B: the same AMS-versus-no-shower comparison as Window A is simply amplified by the Perseids.

RMSE is 0.031 versus 0.309; LIPAS tracks the rise with a residual (peak-day site-mean absolute error ≈0.061), while the no-shower flagship reaches ≈0.606 by 4 August. Radiation favors LIPAS (0.0039 versus 0.034).

Hazard	Hybrid	Flagship	Relative change	Outcome
Solar activity	5.09	5.99	−15%	Hybrid better
Radiation (mSv)	0.0039	0.034	−89%	Hybrid better (quiet-window caveat)
Temperature (°C)	25.5	25.5	0%	Tie (shared physics; vs Diviner-style dayside labels)
Moonquakes (day^{-1})	0.695	0.695	0%	Tie (shared physics; site RMSE varies)
Micrometeorite flux ($\times 10^{-15}$)	0.031	0.309	−90%*	Hybrid better (Perseids / AMS 0.90)
Dust ($g\ cm^{-3}$)	0.169	0.169	0%	Tie (shared physics; vs LDEX-style labels)

Table 2. Window B hybrid versus flagship RMSE (22 July–4 August 2026; lunar-day dominant; 42 site-days).

Table 2 shows the same channel-wise pattern as Window A for solar activity, with a larger micrometeorite separation during Perseids (0.031 versus 0.309). Temperature, moonquakes, and dust remain ties. The large radiation percentage is reported with the quiet-window caveat above.



Window B micrometeorite flux (22 July–4 August 2026), study-site mean. Observed = full AMS+MEM3 label; LIPAS = seasonal + 0.90× shower excess; Flagship = MEM3 without showers. LIPAS tracks Perseids with a visible residual; it is not a perfect overlay.

The Perseids figure shows why micrometeorite accuracies change in Window B: the no-shower MEM 3 baseline stays near the early-window reference then underpredicts into 4 August, while LIPAS follows AMS timing with a conservative coupling residual. This is improvement from including meteor-shower activity versus a seasonal MEM 3 baseline, not a claim of superiority over the full NASA micrometeorite framework.

The micrometeorite result provides evidence that explicit meteor-shower modulation can improve predictions relative to a seasonal MEM 3 baseline during an active shower period. Because the comparison used a MEM 3 baseline without the additional shower modulation, this result is not treated as evidence of general superiority over the complete NASA micrometeorite modeling framework.

The results across both windows show the same overall pattern. Physics matching held for temperature, moonquakes, and dust. Machine learning helps solar activity and radiation predictions in night and day. Temperature, moonquakes, and dust match the flagship physics path as ties. Micrometeorites already favor LIPAS in Window A under AMS coupling; Window B amplifies that same gap as Perseids rise.

5.4 Operational Mission Simulations

A radiation-driven extravehicular-activity (EVA) simulation was executed to test whether the unified hybrid dashboard produces earlier and more actionable abort guidance than consulting public NASA and NOAA products separately during a storm. The simulation used the locked second-train hybrid model at Apollo 11 and an EVA duration equal to Apollo 17 (longest EVA conducted on the moon) EVA-3, 7 hours and 37 minutes. Quiet baseline site drivers from 12 July 2026 were advanced in five-minute steps across the full EVA. Into that quiet baseline, a documented controlled storm script was injected so both LIPAS and the NASA-side methods faced identical situations. At T+0 the crews EVA began under quiet conditions. At T+40 minutes a solar and coronal-mass-ejection-style precursor began to rise, representing the kind of DONKI event context an operator would see before particles arrive. At T+55 minutes geomagnetic activity increased through rising Kp and southward IMF Bz. At T+70 minutes a GOES-like proton flux surge began, representing the Earth-orbit energetic-particle product. The storm then peaked and decayed through the remainder of the EVA.

Identical radiation caution and no-go thresholds were applied to LIPAS and to the NASA physics radiation replica, with caution at 0.095 mSv/h and no-go at 0.15 mSv/h, matching the operational hazard bands already used in the LIPAS mission-planning stack. Separately, a NASA GOES proton alert path used only the proton-flux product, with caution and no-go crossings defined on that product alone. A combined NASA call took the worse of the physics-radiation status and the GOES proton status at each step, approximating an operator who watches both lunar-mapped radiation physics and the live GOES proton stream. LIPAS issued its first caution at T+45 minutes, five minutes after the solar precursor began and twenty-five minutes before the GOES proton surge. This earlier LIPAS warning reflects the different information available to the two pathways. LIPAS incorporated upstream solar and geomagnetic drivers into its radiation estimate, allowing the model to respond to the simulated precursor conditions before the GOES proton-flux threshold was crossed. The result therefore represents earlier warning relative to a GOES-only proton-monitoring pathway, rather than evidence that LIPAS detects radiation events that GOES cannot detect.

LIPAS issued its first caution at T+45 minutes and its no-go at T+85 minutes. The NASA GOES proton product alone gave the caution at T+80 minutes and no-go at T+105 minutes. The NASA physics radiation replica, which detects the precursor drivers before the proton product spikes, reached caution at T+55 minutes and no-go at T+70 minutes. Consequently the combined NASA stack, which is the fairest “what would an attentive NASA-product operator do” baseline, cautioned at T+55 minutes and called no-go at T+70 minutes. LIPAS therefore warned earlier than GOES-only monitoring, which is the comparison most relevant to a fragmented workflow that waits on proton products, while the combined NASA stack still produced an earlier hard no-go in this particular injection because the proton surge is directly included in the physics-based radiation calculation.

This scenario was intentionally radiation-focused. Radiation is an important EVA abort factor, and the first fourteen-day study found hybrid radiation skill against the public NASA/NOAA replica. The mission simulation is not offered as proof that LIPAS is superior on every hazard,.

Additionally, the earlier combined-NASA no-go assumes an operator is actively watching both the lunar-mapped physics radiation path and the live GOES proton product at once. In real EVA workflows those are often sequential, incomplete, or too late to act on as separate products, which is the operational gap a unified dashboard is meant to close.

System	First CAUTION	First NO-GO / return
LIPAS hybrid	T+45	T+85
NASA GOES	T+80	T+105
NASA (combined physics rad + GOES)	T+55	T+70

Overview of the mission simulation results

5.5 Ablation Study: Disabling Cross-Hazard Inputs

This ablation experiment systematically removed hazard channels and driver inputs to quantify their contribution to hybrid model performance.

A short ablation test was run on the same Window A site days to check cross hazard links for solar activity and radiation. The full hybrid model was compared with versions where one hazard input was removed and replaced with that hazard's Window A mean value. The same idea was then repeated with two hazard inputs removed at once. Skill was scored as RMSE on solar activity and on radiation. The goal was not to retune the model. The goal was to see which other hazards interact with eachother most when those two channels are forecast.

Config	Radiation RMSE	Change vs full	Percent worse
Full model (all hazards)	0.1020	0.0000	0.0%
Remove solar	0.1050	-0.0029	-2.9%

Remove radiation	0.1029	-0.0009	-0.9%
Remove temperature	0.1020	~0	~0%
Remove moonquakes	0.1020	~0	~0%
Remove micrometeorites	0.1020	~0	~0%
Remove dust	0.1020	~0	~0%
Remove solar + radiation	0.1068	-0.0048	-4.7%

Table 4 focuses on radiation. Removing the solar activity input made radiation RMSE worse by about 2.9 percent (0.1020 to 0.1050). That was the clearest one hazard effect. Removing radiation itself made only a small change. Removing temperature, moonquakes, micrometeorites, or dust barely moved radiation RMSE. When solar and radiation were both removed, radiation RMSE fell further to 0.1068, about 4.7 percent worse than the full model. This supports a real solar to radiation link in the hybrid.

Table 5 focuses on solar activity. Among the other hazards, removing micrometeorites hurt solar RMSE the most, about 2.2 percent worse (3.899 to 3.984) which was unexpected. Removing radiation or moonquakes also made solar a little worse. Temperature and dust removals were smaller. When the solar input itself was replaced with the Window A mean (from 14 day accuracy studies), solar RMSE got better, not worse. That means the live solar input state was not always helping the blended solar forecast in this night window. The physics heavy path handled solar more cleanly without that noisy input.

Config	Solar RMSE	Change vs full	Percent worse
Full model (all hazards)	3.899	0.000	0.0%
Remove micrometeorites	3.984	-0.085	-2.2%
Remove radiation	3.933	-0.034	-0.9%
Remove moonquakes	3.931	-0.032	-0.8%

Remove temperature	3.916	-0.018	-0.5%
Remove dust	3.909	-0.011	-0.3%
Remove solar	3.290	+0.608	+15.6% (better)
Remove moonquakes + dust	3.960	-0.061	-1.6%

Overall the ablation study proved that radiation skill depends most on keeping solar context. Solar skill depends more weakly on the other hazard channels, with micrometeorites and radiation as the clearest helpers among them. It also proved that removing some of the messy solar data could make LIPAS more accurate in future enhancements.

6. Discussion

The three research questions were addressed through the development of LIPAS, the operational evaluation of its integrated structure, and the quantitative comparison of its predictive performance. The results show that the usefulness of machine learning, physics-based modelling, and data integration depends on the hazard being forecast and on the purpose of the system.

The first research question asked how LIPAS should be developed and what could be learned from the limitations of its original prototype. The development process answered this by showing that machine learning was not equally useful for all six hazards. The original prototype treated each hazard primarily as a machine-learning problem, but this produced unnecessary variation in hazards that were already strongly constrained by physical relationships. These findings led to the final hybrid architecture, in which physics remains dominant where it provides a strong foundation, while machine learning has a greater role where it can provide additional predictive information. The development phase therefore answered the first research question by showing that a hazard-specific hybrid approach was more appropriate than applying machine learning equally to every hazard.

The second research question asked whether integrating multiple lunar environmental hazards into one system could improve operational usability during a simulated mission. The qualitative comparison and mission simulation provided the answer. Existing NASA and NOAA models and datasets remain more mature in terms of observational history, scientific validation, and specialised development, but they generally provide information through separate products. LIPAS brought these different sources and hazards into one forecasting and decision-support environment. During the mission simulation, LIPAS provided earlier caution than the

GOES-only pathway because it incorporated upstream solar and geomagnetic information, although the combined NASA pathway still reached the final no-go condition earlier. The second research question was therefore answered partially but positively: LIPAS demonstrated that integrating multiple hazards can improve how information is presented and used operationally, but it did not consistently produce earlier or safer decisions than a carefully monitored combination of existing products.

Finally, the third research question asked whether the final LIPAS hybrid system could improve prediction performance compared with established physics-based and replicated NASA and NOAA reference models. The quantitative evaluation answered this by showing measurable improvements in selected channels. Solar activity and radiation showed lower RMSE than their replicated flagship baselines in both evaluation windows, while temperature, moonquakes, and dust remained aligned with their physics-based references. Micrometeorite predictions also improved when explicit meteor-shower timing was included, although this result represents an improvement over the specific seasonal MEM 3 comparison used in the study rather than proof of superiority over the complete NASA modelling framework. The third research question was therefore answered positively but selectively: LIPAS improved prediction performance for some hazards while preserving established physical behaviour for others, rather than producing a universal improvement across all six hazards.

The development results are particularly important for interpreting the final architecture. The lack of improvement in temperature, moonquakes, and dust may be viewed as a weakness. However, these channels were deliberately kept mainly physics-led because their underlying relationships were already well represented by the physical models used by LIPAS. Matching the established physics paths therefore shows that the hybrid system did not introduce unnecessary neural corrections into these forecasts. The purpose of the architecture was not to force machine learning into every part of the system, but to allow different sources of information to have different levels of influence.

This is also the main reason the uncertainty-guided weighting is important. LIPAS does not simply average a machine-learning prediction with a physics estimate. The contribution of the learned component changes according to its estimated uncertainty and the environmental conditions. When the neural prediction becomes less certain, its influence is reduced and the physics-based path becomes more important. This allows the system to retain a physically grounded prediction rather than relying entirely on a learned relationship when the model has less confidence in its output.

The strongest evidence for additional predictive information from machine learning came from solar activity and radiation. Solar activity showed improvement over the replicated flagship baseline in both evaluation windows, rather than only during one particular lunar regime.

Radiation also showed lower error, and the ablation experiment provides supporting evidence for why this occurred: removing solar activity information increased radiation error. This suggests that information about solar conditions can contribute to radiation forecasting when it is combined with the physical radiation pathway.

At the same time, the radiation result needs to be interpreted carefully. The evaluation targets are partly based on estimated or physics-derived values rather than continuous co-located measurements from lunar surface radiation instruments (due to missing baseline data). The comparison also used a replicated public baseline rather than a proprietary operational NASA system. The result therefore demonstrates improved predictive behaviour within the experimental framework, but it does not establish that LIPAS measures actual lunar radiation more accurately than a spacecraft instrument.

The mission simulation provides a useful example of why predictive performance and operational usefulness are not exactly the same thing. LIPAS produced an earlier caution than the GOES proton-only pathway because its forecast used upstream solar and geomagnetic information before the simulated proton increase crossed the GOES threshold. However, the combined NASA physics and GOES pathway reached the no-go condition earlier than LIPAS. The result therefore should not be interpreted as LIPAS detecting a storm that NASA could not detect. Instead, it shows that different forecasting pathways can produce different warning times depending on which signals they monitor, how quickly they detect them, and how those signals are converted into operational thresholds. Overall, the advantages of LIPAS lie in its ability to combine several types of information into one warning pathway and provide an earlier caution than the GOES-only workflow. The combined NASA pathway, however, produced the earlier hard no-go decision. Operational usefulness therefore depends not only on forecast accuracy, but also on how forecasts are converted into warnings and decisions.

The micrometeorite results demonstrate another type of improvement. During the Perseid period, LIPAS included explicit meteor-shower timing through AMS-based modulation, while the MEM 3 comparison used the seasonal baseline without that additional shower term. As the Perseids became more active, the two predictions separated substantially, with LIPAS following the additional shower activity more closely. The difference between the quieter Window A and the Perseid-dominated Window B supports this interpretation: when the additional shower contribution was small, the two approaches remained closer, while the difference became larger as the shower approached its peak. This provides evidence that explicitly representing meteor-shower timing can improve predictions relative to a seasonal MEM 3 baseline during an active shower period. However, the experiment did not demonstrate superiority over the complete NASA micrometeorite modelling framework, nor did it provide independent validation against measured lunar impact rates. Instead, it demonstrated the effect of

adding shower timing to a seasonal background model.

The ablation study provides further evidence for the cross hazard dependence inside LIPAS. Removing solar activity information increased radiation error, showing that solar context contributes to the radiation prediction. Other hazard removals produced much smaller changes, suggesting that their influence on radiation was limited under the conditions tested. The solar results were more unexpected. Replacing the live solar input with its Window A mean actually reduced solar error, indicating that the current solar input was not always beneficial under the relatively quiet conditions of that evaluation window.

Several explanations are possible for this ablation solar result. Short-term variation in the live solar input may not match the smoother evaluation targets, the current weighting may give that variable too much influence under quiet conditions, or the construction of the evaluation labels may favour a smoother representation of the underlying signal. The result should therefore not be interpreted as evidence that live solar data are generally harmful. In fact, the radiation ablation results show that solar information can still be useful for another hazard. Further testing under stronger solar activity would be needed to determine when the live solar input provides the greatest benefit.

Looking across the full set of results, the main pattern is that different hazards benefit from different forms of information. Temperature, moonquakes, and dust remained closely aligned with their physics-based paths because those paths already provided an solid foundation. Solar activity and radiation showed the clearest evidence that machine learning and cross-hazard information could add predictive value. Micrometeorite forecasting benefited from explicitly including meteor-shower timing rather than from machine learning alone. The system therefore does not behave as six identical machine-learning models. Instead, it acts as a framework that assigns different forms of information different levels of influence depending on the hazard.

This is the more specific significance of the hybrid architecture. Combining physics and machine learning is not itself a new concept. The contribution of LIPAS is the application of hazard-specific weighting, uncertainty information, and cross-hazard inputs within a single lunar forecasting and decision-support system. The results provide preliminary support for this design, while also showing that the architecture should not be treated as a replacement for specialised scientific models or as a fully validated operational forecasting system.

Several limitations affect the strength of these conclusions. First, the quantitative evaluations rely on proxy targets constructed from public observations, physical formulations, and historical or catalog data rather than continuous co-located lunar surface measurements. The results therefore demonstrate comparative behaviour within the evaluation framework rather than direct accuracy against complete lunar surface truth. Second, the flagship comparisons are replications of publicly documented

methods and products rather than comparisons with proprietary internal operational systems.

The strongest conclusion is therefore not that LIPAS replaces specialised NASA or NOAA models. Its value lies in providing a common forecasting and decision-support layer in which established physics, live space-weather information, cross-hazard relationships, and uncertainty estimates can be considered together. The results support this concept as a promising direction, while more validation may be needed before full operational usability.

7. Future Work

While LIPAS already provides a unified multi-hazard decision-support system for lunar operations, further work is needed before it can be considered an operationally trusted forecasting tool. The completed nightside evaluation, mission simulation, and solar ablation study have helped clarify where future research should focus. The primary limitation remains the lack of continuous lunar surface measurements across all six hazard channels. Rather than repeating the evaluations already completed, future work should build directly on their findings.

7.1 Improving the Physics-Tied Channels

The nightside evaluation showed that LIPAS and the NASA-based reference models matched on temperature, moonquakes, and dust during Window A, as intended for shared physics channels. Micrometeorites already favored LIPAS under AMS coupling, and Window B amplified that separation during Perseids. Future work should strengthen shower timing and MEM 3 integration rather than treating micrometeorites as a pure machine-learning target.

7.2 Understanding the Solar Ablation Result

The solar ablation experiment identified one area that deserves closer investigation. Removing solar context reduced radiation forecast performance, confirming that solar drivers are an important part of the model. However, replacing the live solar inputs with their average values slightly improved the solar activity score. This suggests that the current integration of live solar data introduces noise during relatively quiet nightside conditions rather than indicating that solar physics should be removed from the system.

7.3 Expanding the Solar and Radiation Models

Solar activity and radiation were the only hazard channels that consistently benefited from the hybrid approach, making them the highest priority for future development. Potential improvements include denser sampling from OMNI, DONKI, and SWPC datasets, improved handling of proton flux events, calibrated uncertainty estimates for radiation alerts, and additional validation

against historical space weather events. The mission simulation also demonstrated that the combined NASA physics and GOES approach can issue an abort recommendation earlier in a controlled scenario. Rather than adjusting thresholds retrospectively, future work should investigate these differences transparently and determine which mapping best reflects operational requirements.

7.4 Stronger Validation Through Measured Data

As more data become available from Artemis missions, lunar landers, rovers, and orbital instruments, estimated labels should gradually be replaced with direct measurements wherever possible. Where measured observations are not yet available, estimated values should continue to be identified clearly. Collaboration with organizations such as NASA, NOAA, ESA, JAXA, and university research groups would also strengthen validation by providing access to additional datasets, operational expertise, and mission-relevant performance criteria. LIPAS should continue to be presented as a decision-support platform operating under incomplete observations rather than as a replacement for established scientific instruments or engineering models.

7.5 Extending the Framework Beyond the Lunar Surface

Because much of the system already relies on heliophysics data, extending the framework to cislunar space and Earth orbit is a logical next step (possibly Mars later). These environments offer substantially more observational data than the lunar surface, allowing more rigorous validation of the hybrid methodology. The underlying multi-hazard architecture and channel-specific weighting be reused without requiring a complete redesign.

7.6 Operational Development

Future development should also improve the operational dashboard and EVA planning tools. Clearer uncertainty visualizations, mission-specific alert thresholds, and reports aligned with existing operational checklists would make forecasts easier to interpret during mission planning. These interface improvements are valuable only if they communicate the underlying scientific results accurately and transparently.

7.7 Conclusion

Overall, the immediate priorities are to strengthen the physics-tied hazard channels, resolve the behaviour observed in the solar ablation study, and further validate the hybrid solar and radiation models. Over the medium term, expanding the availability of measured data and working with space agencies will improve confidence in the system. Longer term, the same framework could support cislunar operations and, eventually, Mars missions. Throughout its development, LIPAS should remain positioned as a hybrid decision-support platform designed to operate under incomplete observational data rather than as a universal replacement for specialized scientific models or flight-qualified systems.

8. Conclusion

This research shows that integrating machine learning into a unified, physics-based lunar forecasting system can improve predictive performance while maintaining the physical foundations of established environmental models. The development process answered the first research question by showing that a hazard-specific hybrid approach is more appropriate than applying machine learning uniformly across all hazards. The operational evaluation answered the second by showing that integrating multiple hazards and data sources can improve how information is presented and used during mission planning, although it did not show that LIPAS is universally safer than existing approaches. The quantitative evaluation answered the third by showing measurable improvements in solar activity and radiation, while temperature, moonquakes, and dust remained aligned with their physics-based references. Micrometeorite predictions also improved when meteor-shower timing was explicitly included. Overall, the results support the central research question: a carefully constrained hybrid architecture can add predictive capability where machine learning is useful without replacing physics-based methods where they remain more appropriate.

The main contribution of LIPAS is therefore not the replacement of existing lunar models, but their integration into a single forecasting and decision-support framework. By combining established physical models, public space-weather data, machine-learning inference, and cross-hazard information within one system, LIPAS provides an approach for extending existing scientific capabilities toward operational multi-hazard forecasting. The evaluation also demonstrates the importance of retaining physical constraints when applying machine learning to environmental systems where reliable deterministic relationships already exist.

These findings should be interpreted within the limits of the present evaluation, particularly the absence of continuous in-situ measurements for all six hazards and the use of reproducible proxy and physics-based reference targets. Further validation with future lunar surface measurements, longer evaluation periods, and additional active space-weather conditions will be needed to determine how well the approach generalizes to operational lunar missions. Nevertheless, the results provide evidence that a carefully constrained hybrid architecture can add predictive capability without sacrificing the physical basis of the underlying models.

As programs return to the lunar surface, the ability to anticipate environmental shifts across hazards will matter as much as measuring any one of them well. This project shows how open scientific datasets, a reproducible hybrid forecast stack, and cross-hazard correlation can support safer lunar operations and contribute to the broader future of human space exploration.

Acknowledgements

I would like to thank my mentor, Marjo Lipponen-Mraz, for her guidance, and Benjamin Irani-Fey for his coding support during the NASA Space Apps Challenge. I also thank NASA, NOAA, and other space agencies and research organizations for providing publicly available lunar mission data and space weather archives that made this project possible. I would also like to thank my Polygence research mentor Kyle. K for his rigorous support in helping redesign and re-evaluate the studies, methods and conclusions.

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LIPAS Appendices

Supporting numerical data tables, qualitative classification rules, and reproduction notes for the LIPAS research paper.

Appendix A: Full Numerical Data Tables for Prototype Studies

The tables below are from the 14-day daytime study at 0 degrees N 0 degrees E, April 5 to April 19.

Note: Appendix A tables are from the early prototype study and are not primary accuracy evidence for the current hybrid backend.

Study 1 Predictions (Apr 5–19)

Temperature

Date	LIPAS	Flagship (Diviner)	LRO Observed	LIPAS Error	Diviner Error
Apr 5	-60	-64.40	-68	8.0	3.6
Apr 6	18	14.40	11	7.0	3.4
Apr 7	57	54.10	51	6.0	3.1

Apr 8	84	80.10	77	7.0	3.1
Apr 9	101	98.10	95	6.0	3.1
Apr 10	113	109.9	107	6.0	2.9
Apr 11	119	116.8	115	4.0	1.8
Apr 12	121	119.0	122	1.0	3.0
Apr 13	118	116.8	119	1.0	2.2
Apr 14	112	109.9	111	1.0	1.1
Apr 15	100	98.10	97	3.0	1.1
Apr 16	82	80.10	79	3.0	1.1
Apr 17	56	54.10	52	4.0	2.1
Apr 18	16	14.40	12	4.0	2.4
Apr 19	-62	-64.40	-68	6.0	3.6

Radiation

Date	LIPAS	Flagship (GOES-18)	Observed	LIPAS Error	Flagship Error
Apr 5	0.061	0.058	0.057	0.002	0.000
Apr 6	0.062	0.059	0.058	0.001	0.001
Apr 7	0.064	0.060	0.059	0.001	0.002
Apr 8	0.089	0.078	0.061	0.001	0.004
Apr 9	0.095	0.082	0.066	0.001	0.009
Apr 10	0.072	0.065	0.068	0.001	0.011
Apr 11	0.068	0.063	0.069	0.002	0.012
Apr 12	0.066	0.062	0.082	0.001	0.025
Apr 13	0.065	0.061	0.064	0.001	0.007
Apr 14	0.064	0.060	0.062	0.001	0.005
Apr 15	0.063	0.059	0.091	0.001	0.034
Apr 16	0.062	0.058	0.164	0.009	0.107
Apr 17	0.061	0.057	0.098	0.003	0.041
Apr 18	0.060	0.057	0.070	0.001	0.013
Apr 19	0.059	0.056	0.065	0.001	0.008

Micrometeorites (e-15 /m²/s)

Date	LIPAS	Flagship (MEM 3)	Observed	LIPAS Error	Flagship Error
Apr 5	1.630	1.610	1.612	0.003	0.002
Apr 6	1.640	1.620	1.619	0.003	0.001
Apr 7	1.660	1.630	1.626	0.003	0.004
Apr 8	1.680	1.640	1.637	0.003	0.003
Apr 9	1.720	1.650	1.645	0.002	0.005
Apr 10	1.680	1.650	1.650	0.002	0.000
Apr 11	1.670	1.640	1.643	0.003	0.003
Apr 12	1.670	1.640	1.651	0.002	0.011
Apr 13	1.680	1.650	1.659	0.002	0.009
Apr 14	1.680	1.650	1.656	0.002	0.006

Apr 15	1.690	1.660	1.663	0.002	0.003
Apr 16	1.710	1.670	1.671	0.002	0.001
Apr 17	1.730	1.680	1.682	0.002	0.002
Apr 18	1.740	1.690	1.693	0.002	0.003
Apr 19	1.760	1.700	1.712	0.002	0.012

Moonquakes

Date	LIPAS	Flagship (PSE catalog)	Observed	LIPAS Error	Flagship Error
Apr 5	28	26	26	1	0
Apr 6	29	27	27	0	0
Apr 7	30	27	27	0	0
Apr 8	32	28	27	1	1
Apr 9	31	28	28	0	0
Apr 10	30	28	28	0	0
Apr 11	29	28	27	0	1
Apr 12	28	27	28	0	1
Apr 13	29	27	27	0	0
Apr 14	30	28	28	0	0
Apr 15	30	28	28	0	0
Apr 16	29	28	27	0	1
Apr 17	28	27	28	0	1
Apr 18	28	27	27	0	0
Apr 19	27	26	26	0	0

Dust density

Date	LIPAS	Flagship (LDEX)	Observed	LIPAS Error	Flagship Error
Apr 5	1.550	1.520	1.522	0.005	0.002
Apr 6	1.580	1.540	1.531	0.004	0.009
Apr 7	1.600	1.550	1.543	0.005	0.007
Apr 8	1.640	1.570	1.562	0.004	0.008
Apr 9	1.680	1.590	1.578	0.003	0.012
Apr 10	1.700	1.600	1.594	0.003	0.006
Apr 11	1.730	1.620	1.609	0.002	0.011
Apr 12	1.740	1.630	1.625	0.002	0.005
Apr 13	1.720	1.620	1.618	0.002	0.002
Apr 14	1.710	1.610	1.607	0.002	0.003
Apr 15	1.680	1.600	1.598	0.002	0.002
Apr 16	1.650	1.580	1.583	0.002	0.003
Apr 17	1.620	1.560	1.561	0.002	0.001
Apr 18	1.590	1.540	1.542	0.002	0.002
Apr 19	1.560	1.520	1.524	0.002	0.004

Solar activity

Date	LIPAS	Flagship (NOAA Kp)	Observed	LIPAS Error	Flagship Error
Apr 5	4.500	4.200	3.1	0.1	0.1
Apr 6	4.800	4.500	3.3	0.1	0.1
Apr 7	5.100	4.800	3.4	0.1	0.1
Apr 8	8.900	8.200	3.5	0.0	0.0
Apr 9	9.400	8.700	4.0	0.0	0.0
Apr 10	5.700	5.300	4.1	0.0	0.0
Apr 11	4.900	4.600	4.2	0.1	0.1
Apr 12	4.400	4.100	5.1	0.1	0.1
Apr 13	4.100	3.800	3.9	0.1	0.1
Apr 14	4.000	3.700	3.8	0.0	0.0
Apr 15	3.900	3.600	5.5	0.0	0.0
Apr 16	3.800	3.500	8.3	0.7	0.7
Apr 17	3.700	3.400	6.2	0.1	0.1
Apr 18	3.600	3.300	3.7	0.0	0.0
Apr 19	3.500	3.200	3.8	0.0	0.0

Appendix B: Full Numerical Data Tables

The tables below are from the 14-day daytime study at 0 degrees N 0 degrees E, April 5 to April 19.

Study 2 Predictions (Apr 5–19)

Date	Diviner formula (°C)	LIPAS (°C)	LRO observed (°C)	LIPAS error (°C)	Diviner error (°C)
Apr 5	-64.4	-60	-68	8	4
Apr 6	14.4	18	11	7	3
Apr 7	54.1	57	51	6	3
Apr 8	80.1	84	77	7	3
Apr 9	98.1	101	95	6	3
Apr 10	109.9	113	107	6	3
Apr 11	116.8	119	115	4	2
Apr 12	119.0	121	122	1	3
Apr 13	116.8	118	119	1	2
Apr 14	109.9	112	111	1	1
Apr 15	98.1	100	97	3	1
Apr 16	80.1	82	79	3	1
Apr 17	54.1	56	52	4	2
Apr 18	14.4	16	12	4	2
Apr 19	-64.4	-62	-68	6	4

Appendix C: Qualitative Comparison Framework

Classification rules for each criteria can be found below:

Hazard Coverage: None = no hazard addressed. Very Low = one hazard type only. Low = up to two hazards. Moderate = three to four hazards. High = five hazards. Very High = six or more hazards integrated.

Data Fusion Capability: None = single dataset only. Low = multiple datasets with no integration. Moderate = partial integration of multiple sources. High = integrated multi-source system. Very High = real-time dynamic multi-source fusion.

Temporal Forecasting Ability: None = no forecasting, archival only. Low = event-driven alerts, no time horizon. Moderate = short-range estimates for narrow domain. High = multi-day forecasting for one or more hazards. Very High = multi-day forecasting across all hazards simultaneously.

Spatial Coverage: None = no spatial resolution. Very Low = limited to specific historical measurement sites. Low = single region or polar area only. Moderate = broad regional coverage. High = global or near-global coverage of relevant environments. Very High = complete global lunar surface coverage with location selection.

Real-time Operational Capability: None = mission ended or historical only. Low = archival access with significant delay. Moderate = near-real-time with limited update frequency. High = live data feeds with regular updates. Very High = continuous real-time data with automated processing.

Robustness and Stability: Low = limited coverage or known gaps. Moderate = functional with notable dependencies on external APIs or data sources. High = well-validated and operationally stable within a narrow domain. Very High = continuously calibrated mission-grade instruments with long operational history.

Validation Strength: None = no comparison against observed data. Low = informal or limited comparison. Moderate = structured comparison against observed data for a limited period. High = peer-reviewed validation against direct measurements. Very High = direct physical measurement, self-validating by design.

Usability and Operational Readiness: Very Low = raw scientific data only, requires specialist processing. Low = accessible via technical interfaces for trained users. Moderate = accessible via API or web portal with some technical knowledge. High = clear interface for operational use. Very High = interactive dashboard designed for non-specialist mission planners.

Openness and Accessibility: Low = restricted access or proprietary. Moderate = available via data archives with registration or processing requirements. High = publicly available via open API or published datasets. Very High = fully open, browser-based, no setup required.

Appendix D: Window A Full Numerical Data Tables

The tables below are from the Window A fourteen-day lunar-night study, 5 July to 18 July 2026, at three sites: equatorial reference 0°N 0°E, Apollo 11, and the Artemis III Nobile Rim candidate (forty-two site-days).
Note: Appendix D tables are from the final hybrid Window A study. “Observed / label” values are next-step extended-dataset labels (public drivers plus physics and catalog anchors).

Format: Date, LIPAS, Flagship, Observed / label, LIPAS Error, Flagship Error. Error = absolute difference from the label.

Site: Equatorial reference (0°N, 0°E)

Solar activity

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	19.81	25.43	8.78	11.04	16.65
Jul 6	9.92	7.01	7.34	2.58	0.34
Jul 7	9.60	6.22	7.71	1.89	1.49
Jul 8	9.61	6.57	7.66	1.94	1.09
Jul 9	9.80	6.66	8.05	1.75	1.39
Jul 10	9.94	6.87	7.85	2.09	0.98
Jul 11	9.88	6.76	8.53	1.35	1.77
Jul 12	10.17	7.13	7.84	2.33	0.71
Jul 13	10.14	6.75	7.84	2.29	1.09
Jul 14	9.78	6.76	8.11	1.67	1.35
Jul 15	10.18	6.90	7.37	2.81	0.47
Jul 16	9.71	6.50	7.44	2.27	0.95
Jul 17	10.00	6.54	8.25	1.75	1.71
Jul 18	10.14	6.98	3.83	6.30	3.14

Radiation (mSv)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	0.1962	0.0822	0.1770	0.0192	0.0948
Jul 6	0.0694	0.0324	0.1643	0.0949	0.1319
Jul 7	0.0695	0.0352	0.1670	0.0975	0.1318
Jul 8	0.0694	0.0374	0.1627	0.0933	0.1252
Jul 9	0.0694	0.0390	0.1610	0.0916	0.1220
Jul 10	0.0695	0.0382	0.3146	0.2451	0.2764
Jul 11	0.0809	0.0386	0.1974	0.1165	0.1588
Jul 12	0.0694	0.0374	0.1537	0.0842	0.1163
Jul 13	0.0695	0.0387	0.1517	0.0822	0.1130
Jul 14	0.0694	0.0386	0.1543	0.0849	0.1157
Jul 15	0.0694	0.0381	0.1533	0.0839	0.1152
Jul 16	0.0695	0.0397	0.1567	0.0872	0.1170

Jul 17	0.0695	0.0395	0.0885	0.0190	0.0490
Jul 18	0.0694	0.0379	0.0615	0.0079	0.0237

Temperature (°C)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	80.7	80.7	80.7	0.0	0.0
Jul 6	56.7	56.7	56.7	0.0	0.0
Jul 7	20.9	20.9	20.9	0.0	0.0
Jul 8	-43.6	-43.6	-43.6	0.0	0.0
Jul 9	-29.8	-29.8	-29.8	0.0	0.0
Jul 10	-50.4	-50.4	-50.4	0.0	0.0
Jul 11	-68.7	-68.7	-68.7	0.0	0.0
Jul 12	-84.9	-84.9	-84.9	0.0	0.0
Jul 13	-99.2	-99.2	-99.2	0.0	0.0
Jul 14	-111.7	-111.7	-111.7	0.0	0.0
Jul 15	-122.6	-122.6	-122.6	0.0	0.0
Jul 16	-132.0	-132.0	-132.0	0.0	0.0
Jul 17	-140.2	-140.2	-140.2	0.0	0.0
Jul 18	-147.2	-147.2	-147.2	0.0	0.0

Moonquakes (per day)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	49.02	49.02	49.02	0.00	0.00
Jul 6	51.05	51.05	51.05	0.00	0.00
Jul 7	52.16	52.16	52.16	0.00	0.00
Jul 8	51.55	51.55	51.55	0.00	0.00
Jul 9	50.34	50.34	50.34	0.00	0.00
Jul 10	48.31	48.31	48.31	0.00	0.00
Jul 11	46.27	46.27	46.27	0.00	0.00
Jul 12	44.81	44.81	44.81	0.00	0.00
Jul 13	44.10	44.10	44.10	0.00	0.00
Jul 14	43.74	43.74	43.74	0.00	0.00
Jul 15	43.74	43.74	43.74	0.00	0.00
Jul 16	43.74	43.74	43.74	0.00	0.00
Jul 17	44.41	44.41	44.41	0.00	0.00
Jul 18	45.73	45.73	45.73	0.00	0.00

Micrometeorite flux ($\times 10^{-15}$)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	2.2924	2.2924	2.2924	0.0000	0.0000
Jul 6	2.3172	2.3172	2.3172	0.0000	0.0000
Jul 7	2.3338	2.3338	2.3338	0.0000	0.0000

Jul 8	2.3380	2.3380	2.3380	0.0000	0.0000
Jul 9	2.3281	2.3281	2.3281	0.0000	0.0000
Jul 10	2.3060	2.3060	2.3060	0.0000	0.0000
Jul 11	2.2757	2.2757	2.2757	0.0000	0.0000
Jul 12	2.2428	2.2428	2.2428	0.0000	0.0000
Jul 13	2.2119	2.2119	2.2119	0.0000	0.0000
Jul 14	2.1860	2.1860	2.1860	0.0000	0.0000
Jul 15	2.1661	2.1661	2.1661	0.0000	0.0000
Jul 16	2.1518	2.1518	2.1518	0.0000	0.0000
Jul 17	2.1417	2.1417	2.1417	0.0000	0.0000
Jul 18	2.1343	2.1343	2.1343	0.0000	0.0000

Dust (g cm⁻³)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	1.5612	1.5612	1.5612	0.0000	0.0000
Jul 6	1.5959	1.5959	1.5959	0.0000	0.0000
Jul 7	1.6298	1.6298	1.6298	0.0000	0.0000
Jul 8	1.6625	1.6625	1.6625	0.0000	0.0000
Jul 9	1.6664	1.6664	1.6664	0.0000	0.0000
Jul 10	1.6320	1.6320	1.6320	0.0000	0.0000
Jul 11	1.5967	1.5967	1.5967	0.0000	0.0000
Jul 12	1.5612	1.5612	1.5612	0.0000	0.0000
Jul 13	1.5495	1.5495	1.5495	0.0000	0.0000
Jul 14	1.5469	1.5469	1.5469	0.0000	0.0000
Jul 15	1.5449	1.5449	1.5449	0.0000	0.0000
Jul 16	1.5435	1.5435	1.5435	0.0000	0.0000
Jul 17	1.5425	1.5425	1.5425	0.0000	0.0000
Jul 18	1.5418	1.5418	1.5418	0.0000	0.0000

Site: Apollo 11

Solar activity

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	19.86	25.43	8.78	11.08	16.65
Jul 6	10.18	7.01	7.34	2.83	0.34
Jul 7	9.82	6.22	7.71	2.11	1.49
Jul 8	9.84	6.57	7.66	2.18	1.09
Jul 9	9.75	6.66	8.05	1.70	1.39
Jul 10	9.84	6.87	7.85	1.99	0.98
Jul 11	9.75	6.76	8.53	1.23	1.77

Jul 12	10.16	7.13	7.84	2.33	0.71
Jul 13	10.05	6.75	7.84	2.21	1.09
Jul 14	9.95	6.76	8.11	1.84	1.35
Jul 15	10.11	6.90	7.37	2.74	0.47
Jul 16	9.64	6.50	7.44	2.20	0.95
Jul 17	9.75	6.54	8.25	1.50	1.71
Jul 18	10.16	6.98	3.83	6.33	3.14

Radiation (mSv)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	0.1963	0.0822	0.1770	0.0193	0.0948
Jul 6	0.0695	0.0324	0.1643	0.0949	0.1319
Jul 7	0.0695	0.0352	0.1670	0.0975	0.1318
Jul 8	0.0694	0.0374	0.1627	0.0933	0.1252
Jul 9	0.0694	0.0390	0.1610	0.0916	0.1220
Jul 10	0.0695	0.0382	0.3146	0.2451	0.2764
Jul 11	0.0809	0.0386	0.1974	0.1166	0.1588
Jul 12	0.0694	0.0374	0.1537	0.0842	0.1163
Jul 13	0.0695	0.0387	0.1517	0.0821	0.1130
Jul 14	0.0695	0.0386	0.1543	0.0849	0.1157
Jul 15	0.0695	0.0381	0.1533	0.0838	0.1152
Jul 16	0.0694	0.0397	0.1567	0.0872	0.1170
Jul 17	0.0695	0.0395	0.0885	0.0191	0.0490
Jul 18	0.0695	0.0379	0.0615	0.0079	0.0237

Temperature (°C)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	21.2	21.2	-36.7	57.8	57.8
Jul 6	-42.9	-42.9	-28.1	14.8	14.8
Jul 7	-95.4	-95.4	-48.9	46.5	46.5
Jul 8	-129.2	-129.2	-67.4	61.8	61.8
Jul 9	-136.4	-136.4	-83.8	52.6	52.6
Jul 10	-143.1	-143.1	-98.2	45.0	45.0
Jul 11	-148.7	-148.7	-110.8	37.9	37.9
Jul 12	-153.5	-153.5	-121.8	31.7	31.7
Jul 13	-157.7	-157.7	-131.3	26.4	26.4
Jul 14	-161.0	-161.0	-139.6	21.4	21.4
Jul 15	-163.5	-163.5	-146.7	16.8	16.8
Jul 16	-166.4	-166.4	-152.9	13.5	13.5
Jul 17	-169.1	-169.1	-158.3	10.8	10.8
Jul 18	-171.3	-171.3	-163.2	8.1	8.1

Moonquakes (per day)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	50.34	50.34	52.38	2.03	2.03
Jul 6	52.38	52.38	52.99	0.61	0.61
Jul 7	52.99	52.99	50.96	2.03	2.03
Jul 8	50.96	50.96	48.92	2.03	2.03
Jul 9	48.92	48.92	46.89	2.03	2.03
Jul 10	46.89	46.89	44.86	2.03	2.03
Jul 11	44.86	44.86	43.70	1.16	1.16
Jul 12	43.70	43.70	43.70	0.00	0.00
Jul 13	43.70	43.70	43.70	0.00	0.00
Jul 14	43.70	43.70	43.70	0.00	0.00
Jul 15	43.70	43.70	43.70	0.00	0.00
Jul 16	43.70	43.70	44.73	1.03	1.03
Jul 17	44.73	44.73	46.76	2.03	2.03
Jul 18	46.76	46.76	48.79	2.03	2.03

Micrometeorite flux ($\times 10^{-15}$)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	2.2918	2.2918	1.7710	0.5208	0.5208
Jul 6	2.3166	2.3166	1.7690	0.5476	0.5476
Jul 7	2.3332	2.3332	1.7670	0.5662	0.5662
Jul 8	2.3374	2.3374	1.7650	0.5724	0.5724
Jul 9	2.3275	2.3275	1.7630	0.5645	0.5645
Jul 10	2.3054	2.3054	1.7610	0.5443	0.5443
Jul 11	2.2751	2.2751	1.7590	0.5161	0.5161
Jul 12	2.2422	2.2422	1.7571	0.4851	0.4851
Jul 13	2.2113	2.2113	1.7551	0.4562	0.4562
Jul 14	2.1854	2.1854	1.7531	0.4322	0.4322
Jul 15	2.1655	2.1655	1.7512	0.4143	0.4143
Jul 16	2.1512	2.1512	1.7493	0.4019	0.4019
Jul 17	2.1411	2.1411	1.7473	0.3938	0.3938
Jul 18	2.1338	2.1338	1.7454	0.3883	0.3883

Dust (g cm^{-3})

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	1.6173	1.6173	1.2681	0.3491	0.3491
Jul 6	1.6519	1.6519	0.8841	0.7678	0.7678
Jul 7	1.6632	1.6632	0.8663	0.7969	0.7969
Jul 8	1.6314	1.6314	0.8485	0.7829	0.7829
Jul 9	1.5984	1.5984	0.8308	0.7676	0.7676
Jul 10	1.5640	1.5640	0.8246	0.7394	0.7394
Jul 11	1.5501	1.5501	0.8245	0.7256	0.7256
Jul 12	1.5468	1.5468	0.8244	0.7224	0.7224

Jul 13	1.5438	1.5438	0.8243	0.7195	0.7195
Jul 14	1.5412	1.5412	0.8242	0.7170	0.7170
Jul 15	1.5393	1.5393	0.8639	0.6753	0.6753
Jul 16	1.5378	1.5378	1.0293	0.5085	0.5085
Jul 17	1.5368	1.5368	1.2103	0.3266	0.3266
Jul 18	1.5450	1.5450	1.4936	0.0514	0.0514

Site: Artemis III Nobile Rim candidate

Solar activity

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	19.88	25.43	8.78	11.11	16.65
Jul 6	10.08	7.01	7.34	2.74	0.34
Jul 7	9.59	6.22	7.71	1.89	1.49
Jul 8	9.94	6.57	7.66	2.27	1.09
Jul 9	9.86	6.66	8.05	1.81	1.39
Jul 10	9.94	6.87	7.85	2.08	0.98
Jul 11	9.96	6.76	8.53	1.43	1.77
Jul 12	10.04	7.13	7.84	2.20	0.71
Jul 13	9.91	6.75	7.84	2.06	1.09
Jul 14	9.85	6.76	8.11	1.74	1.35
Jul 15	9.79	6.90	7.37	2.41	0.47
Jul 16	9.53	6.50	7.44	2.09	0.95
Jul 17	9.80	6.54	8.25	1.55	1.71
Jul 18	10.06	6.98	3.83	6.23	3.14

Radiation (mSv)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	0.2132	0.0881	0.1770	0.0362	0.0889
Jul 6	0.0694	0.0348	0.1643	0.0949	0.1296
Jul 7	0.0695	0.0378	0.1670	0.0975	0.1292
Jul 8	0.0695	0.0401	0.1627	0.0931	0.1225
Jul 9	0.0695	0.0418	0.1610	0.0915	0.1192
Jul 10	0.0695	0.0410	0.3146	0.2452	0.2737
Jul 11	0.0868	0.0414	0.1974	0.1106	0.1560
Jul 12	0.0694	0.0400	0.1537	0.0843	0.1136
Jul 13	0.0694	0.0414	0.1517	0.0822	0.1102
Jul 14	0.0694	0.0414	0.1543	0.0849	0.1129
Jul 15	0.0694	0.0409	0.1533	0.0839	0.1125
Jul 16	0.0694	0.0425	0.1567	0.0873	0.1141

Jul 17	0.0694	0.0424	0.0885	0.0191	0.0462
Jul 18	0.0695	0.0406	0.0645	0.0049	0.0240

Temperature (°C)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	-107.4	-107.4	-204.8	97.4	97.4
Jul 6	-112.8	-112.8	-99.7	13.2	13.2
Jul 7	-104.9	-104.9	-120.5	15.5	15.5
Jul 8	-111.4	-111.4	-139.0	27.6	27.6
Jul 9	-124.3	-124.3	-155.3	31.0	31.0
Jul 10	-132.8	-132.8	-169.7	36.9	36.9
Jul 11	-134.4	-134.4	-182.4	47.9	47.9
Jul 12	-139.2	-139.2	-193.3	54.2	54.2
Jul 13	-151.5	-151.5	-202.9	51.4	51.4
Jul 14	-159.8	-159.8	-211.1	51.3	51.3
Jul 15	-153.1	-153.1	-218.3	65.2	65.2
Jul 16	-157.7	-157.7	-224.5	66.8	66.8
Jul 17	-159.1	-159.1	-229.9	70.8	70.8
Jul 18	-155.6	-155.6	-230.0	74.4	74.4

Moonquakes (per day)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	47.38	47.38	49.42	2.03	2.03
Jul 6	49.42	49.42	50.03	0.61	0.61
Jul 7	50.03	50.03	47.99	2.03	2.03
Jul 8	47.99	47.99	45.96	2.03	2.03
Jul 9	45.96	45.96	43.93	2.03	2.03
Jul 10	43.93	43.93	41.90	2.03	2.03
Jul 11	41.90	41.90	40.74	1.16	1.16
Jul 12	40.74	40.74	40.74	0.00	0.00
Jul 13	40.74	40.74	40.74	0.00	0.00
Jul 14	40.74	40.74	40.74	0.00	0.00
Jul 15	40.74	40.74	40.74	0.00	0.00
Jul 16	40.74	40.74	41.77	1.03	1.03
Jul 17	41.77	41.77	43.80	2.03	2.03
Jul 18	43.80	43.80	45.83	2.03	2.03

Micrometeorite flux (×10⁻¹⁵)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	2.2138	2.2138	1.7069	0.5069	0.5069
Jul 6	2.2378	2.2378	1.7050	0.5328	0.5328
Jul 7	2.2538	2.2538	1.7030	0.5508	0.5508

Jul 8	2.2579	2.2579	1.7011	0.5567	0.5567
Jul 9	2.2484	2.2484	1.6992	0.5491	0.5491
Jul 10	2.2269	2.2269	1.6973	0.5296	0.5296
Jul 11	2.1977	2.1977	1.6954	0.5023	0.5023
Jul 12	2.1659	2.1659	1.6935	0.4724	0.4724
Jul 13	2.1361	2.1361	1.6916	0.4444	0.4444
Jul 14	2.1110	2.1110	1.6897	0.4213	0.4213
Jul 15	2.0919	2.0919	1.6879	0.4040	0.4040
Jul 16	2.0780	2.0780	1.6860	0.3920	0.3920
Jul 17	2.0683	2.0683	1.6841	0.3841	0.3841
Jul 18	2.0612	2.0612	1.6823	0.3789	0.3789

Dust (g cm⁻³)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 5	1.5690	1.5690	1.2310	0.3380	0.3380
Jul 6	1.6033	1.6033	0.8583	0.7450	0.7450
Jul 7	1.6145	1.6145	0.8407	0.7738	0.7738
Jul 8	1.5830	1.5830	0.8230	0.7600	0.7600
Jul 9	1.5501	1.5501	0.8053	0.7448	0.7448
Jul 10	1.5161	1.5161	0.7992	0.7168	0.7168
Jul 11	1.5023	1.5023	0.7991	0.7032	0.7032
Jul 12	1.4992	1.4992	0.7990	0.7001	0.7001
Jul 13	1.4962	1.4962	0.7989	0.6973	0.6973
Jul 14	1.4938	1.4938	0.7988	0.6950	0.6950
Jul 15	1.4919	1.4919	0.8373	0.6546	0.6546
Jul 16	1.4905	1.4905	0.9977	0.4929	0.4929
Jul 17	1.4896	1.4896	1.1732	0.3164	0.3164
Jul 18	1.4977	1.4977	1.4486	0.0491	0.0491

Appendix E: Window B Full Numerical Data Tables

The tables below are from the Window B fourteen-day lunar-day study, 22 July to 4 August 2026, at three sites: equatorial reference 0°N 0°E, Apollo 11, and the Artemis III Nobile Rim candidate (forty-two site-days).

Note: Appendix E tables are from the final hybrid Window B study. “Observed / label” values are next-step extended-dataset labels (public drivers plus physics and catalog anchors). They are not continuous Artemis in-situ instruments for every hazard at every coordinate.

Format matches Appendix D: Date, LIPAS, Flagship, Observed / label, LIPAS Error, Flagship Error.

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	2.0906	2.0900	2.0906	0.0000	0.0006
Jul 23	2.0871	2.0854	2.0871	0.0000	0.0018
Jul 24	2.0852	2.0807	2.0852	0.0000	0.0045
Jul 25	2.0864	2.0759	2.0864	0.0000	0.0105
Jul 26	2.0937	2.0710	2.0937	0.0000	0.0228
Jul 27	2.1114	2.0659	2.1114	0.0000	0.0455
Jul 28	2.1444	2.0607	2.1444	0.0000	0.0837
Jul 29	2.1973	2.0554	2.1973	0.0000	0.1419
Jul 30	2.2717	2.0500	2.2717	0.0000	0.2217
Jul 31	2.3637	2.0445	2.3637	0.0000	0.3192
Aug 1	2.4625	2.0389	2.4625	0.0000	0.4236
Aug 2	2.5512	2.0331	2.5512	0.0000	0.5181
Aug 3	2.6111	2.0273	2.6111	0.0000	0.5839
Aug 4	2.6277	2.0213	2.6277	0.0000	0.6064

Site: Equatorial reference (0°N, 0°E)

Solar activity

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	13.31	12.36	18.35	5.04	5.99
Jul 23	13.41	12.36	18.35	4.94	5.99
Jul 24	13.21	12.36	18.35	5.14	5.99
Jul 25	13.23	12.36	18.35	5.12	5.99
Jul 26	13.29	12.36	18.35	5.06	5.99
Jul 27	13.50	12.36	18.35	4.85	5.99
Jul 28	13.46	12.30	18.35	4.89	6.05
Jul 29	13.37	12.28	18.35	4.98	6.07
Jul 30	13.28	12.28	18.35	5.07	6.07
Jul 31	13.13	12.28	18.35	5.22	6.07
Aug 1	12.99	12.28	18.35	5.36	6.07
Aug 2	13.30	12.39	18.35	5.05	5.96
Aug 3	13.32	12.79	18.35	5.03	5.56
Aug 4	13.30	12.28	18.35	5.05	6.07

Radiation (mSv)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	0.0695	0.0385	0.0728	0.0034	0.0343
Jul 23	0.0694	0.0385	0.0728	0.0034	0.0343

Jul 24	0.0694	0.0385	0.0728	0.0034	0.0343
Jul 25	0.0693	0.0385	0.0728	0.0035	0.0343
Jul 26	0.0696	0.0385	0.0728	0.0033	0.0343
Jul 27	0.0694	0.0385	0.0728	0.0034	0.0343
Jul 28	0.0696	0.0373	0.0728	0.0033	0.0356
Jul 29	0.0694	0.0365	0.0728	0.0034	0.0363
Jul 30	0.0694	0.0365	0.0728	0.0035	0.0363
Jul 31	0.0694	0.0365	0.0728	0.0034	0.0363
Aug 1	0.0694	0.0365	0.0728	0.0035	0.0363
Aug 2	0.0695	0.0390	0.0728	0.0034	0.0338
Aug 3	0.0694	0.0440	0.0749	0.0056	0.0309
Aug 4	0.0695	0.0365	0.0728	0.0034	0.0363

Temperature (°C)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	-167.8	-167.8	-167.8	0.0	0.0
Jul 23	-171.9	-171.9	-171.9	0.0	0.0
Jul 24	-35.6	-35.6	-35.6	0.0	0.0
Jul 25	24.7	24.7	24.7	0.0	0.0
Jul 26	59.3	59.3	59.3	0.0	0.0
Jul 27	82.5	82.5	82.5	0.0	0.0
Jul 28	98.4	98.4	98.4	0.0	0.0
Jul 29	108.8	108.8	108.8	0.0	0.0
Jul 30	114.3	114.3	114.3	0.0	0.0
Jul 31	115.3	115.3	115.3	0.0	0.0
Aug 1	111.9	111.9	111.9	0.0	0.0
Aug 2	103.8	103.8	103.8	0.0	0.0
Aug 3	90.5	90.5	90.5	0.0	0.0
Aug 4	70.9	70.9	70.9	0.0	0.0

Moonquakes (per day)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	49.59	49.59	49.59	0.00	0.00
Jul 23	49.84	49.84	49.84	0.00	0.00
Jul 24	49.15	49.15	49.15	0.00	0.00
Jul 25	47.97	47.97	47.97	0.00	0.00
Jul 26	46.80	46.80	46.80	0.00	0.00
Jul 27	45.93	45.93	45.93	0.00	0.00
Jul 28	45.22	45.22	45.22	0.00	0.00
Jul 29	45.04	45.04	45.04	0.00	0.00
Jul 30	45.04	45.04	45.04	0.00	0.00
Jul 31	45.04	45.04	45.04	0.00	0.00
Aug 1	45.38	45.38	45.38	0.00	0.00

Aug 2	45.84	45.84	45.84	0.00	0.00
Aug 3	46.88	46.88	46.88	0.00	0.00
Aug 4	48.06	48.06	48.06	0.00	0.00

Micrometeorite flux (×10⁻¹⁵)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	2.1149	2.1143	2.1149	0.0000	0.0006
Jul 23	2.1115	2.1097	2.1115	0.0000	0.0018
Jul 24	2.1095	2.1049	2.1095	0.0000	0.0045
Jul 25	2.1107	2.1001	2.1107	0.0000	0.0106
Jul 26	2.1181	2.0951	2.1181	0.0000	0.0230
Jul 27	2.1360	2.0900	2.1360	0.0000	0.0460
Jul 28	2.1694	2.0847	2.1694	0.0000	0.0846
Jul 29	2.2229	2.0794	2.2229	0.0000	0.1435
Jul 30	2.2982	2.0739	2.2982	0.0000	0.2243
Jul 31	2.3913	2.0683	2.3913	0.0000	0.3229
Aug 1	2.4912	2.0626	2.4912	0.0000	0.4286
Aug 2	2.5809	2.0568	2.5809	0.0000	0.5241
Aug 3	2.6415	2.0509	2.6415	0.0000	0.5907
Aug 4	2.6583	2.0448	2.6583	0.0000	0.6134

Dust (g cm⁻³)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	1.6157	1.6157	1.6157	0.0000	0.0000
Jul 23	1.6476	1.6476	1.6476	0.0000	0.0000
Jul 24	1.6372	1.6372	1.6372	0.0000	0.0000
Jul 25	1.6050	1.6050	1.6050	0.0000	0.0000
Jul 26	1.5735	1.5735	1.5735	0.0000	0.0000
Jul 27	1.5430	1.5430	1.5430	0.0000	0.0000
Jul 28	1.5453	1.5453	1.5453	0.0000	0.0000
Jul 29	1.5506	1.5506	1.5506	0.0000	0.0000
Jul 30	1.5580	1.5580	1.5580	0.0000	0.0000
Jul 31	1.5673	1.5673	1.5673	0.0000	0.0000
Aug 1	1.5772	1.5772	1.5772	0.0000	0.0000
Aug 2	1.5861	1.5861	1.5861	0.0000	0.0000
Aug 3	1.5921	1.5921	1.5921	0.0000	0.0000
Aug 4	1.6127	1.6127	1.6127	0.0000	0.0000

Site: Apollo 11

Solar activity

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	13.19	12.36	18.35	5.16	5.99
Jul 23	13.46	12.36	18.35	4.89	5.99
Jul 24	13.37	12.36	18.35	4.98	5.99
Jul 25	13.17	12.36	18.35	5.18	5.99
Jul 26	13.07	12.36	18.35	5.28	5.99
Jul 27	13.35	12.36	18.35	5.00	5.99
Jul 28	13.27	12.30	18.35	5.08	6.05
Jul 29	13.32	12.28	18.35	5.03	6.07
Jul 30	13.22	12.28	18.35	5.13	6.07
Jul 31	13.37	12.28	18.35	4.98	6.07
Aug 1	13.38	12.28	18.35	4.97	6.07
Aug 2	13.29	12.39	18.35	5.06	5.96
Aug 3	13.58	12.79	18.35	4.77	5.56
Aug 4	13.29	12.28	18.35	5.06	6.07

Radiation (mSv)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	0.0695	0.0385	0.0728	0.0033	0.0343
Jul 23	0.0694	0.0385	0.0728	0.0034	0.0343
Jul 24	0.0694	0.0385	0.0728	0.0034	0.0343
Jul 25	0.0694	0.0385	0.0728	0.0034	0.0343
Jul 26	0.0694	0.0385	0.0728	0.0034	0.0343
Jul 27	0.0696	0.0385	0.0728	0.0032	0.0343
Jul 28	0.0695	0.0373	0.0728	0.0033	0.0356
Jul 29	0.0694	0.0365	0.0728	0.0034	0.0363
Jul 30	0.0694	0.0365	0.0728	0.0034	0.0363
Jul 31	0.0695	0.0365	0.0728	0.0033	0.0363
Aug 1	0.0695	0.0365	0.0728	0.0034	0.0363
Aug 2	0.0695	0.0390	0.0728	0.0034	0.0338
Aug 3	0.0695	0.0440	0.0749	0.0054	0.0309
Aug 4	0.0695	0.0365	0.0728	0.0033	0.0363

Temperature (°C)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	-79.7	-79.7	-79.7	0.0	0.0
Jul 23	16.6	16.6	16.6	0.0	0.0
Jul 24	51.0	51.0	51.0	0.0	0.0
Jul 25	81.3	81.3	81.3	0.0	0.0
Jul 26	99.9	99.9	99.9	0.0	0.0

Jul 27	110.9	110.9	110.9	0.0	0.0
Jul 28	117.2	117.2	117.2	0.0	0.0
Jul 29	119.8	119.8	119.8	0.0	0.0
Jul 30	115.2	115.2	115.2	0.0	0.0
Jul 31	106.2	106.2	106.2	0.0	0.0
Aug 1	92.2	92.2	92.2	0.0	0.0
Aug 2	69.8	69.8	69.8	0.0	0.0
Aug 3	39.3	39.3	39.3	0.0	0.0
Aug 4	1.9	1.9	1.9	0.0	0.0

Moonquakes (per day)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	50.08	50.08	48.77	1.31	1.31
Jul 23	48.91	48.91	48.06	0.85	0.85
Jul 24	47.73	47.73	47.35	0.38	0.38
Jul 25	46.56	46.56	46.64	0.08	0.08
Jul 26	45.39	45.39	45.93	0.54	0.54
Jul 27	44.99	44.99	45.69	0.70	0.70
Jul 28	44.99	44.99	45.69	0.70	0.70
Jul 29	44.99	44.99	45.69	0.70	0.70
Jul 30	44.99	44.99	45.69	0.70	0.70
Jul 31	44.99	44.99	45.69	0.70	0.70
Aug 1	45.86	45.86	46.21	0.35	0.35
Aug 2	47.03	47.03	46.92	0.11	0.11
Aug 3	48.21	48.21	47.64	0.57	0.57
Aug 4	49.38	49.38	48.35	1.03	1.03

Micrometeorite flux ($\times 10^{-15}$)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	2.1144	2.1137	2.1144	0.0000	0.0006
Jul 23	2.1109	2.1091	2.1109	0.0000	0.0018
Jul 24	2.1089	2.1044	2.1089	0.0000	0.0045
Jul 25	2.1101	2.0995	2.1101	0.0000	0.0106
Jul 26	2.1176	2.0945	2.1176	0.0000	0.0230
Jul 27	2.1354	2.0894	2.1354	0.0000	0.0460
Jul 28	2.1688	2.0842	2.1688	0.0000	0.0846
Jul 29	2.2223	2.0788	2.2223	0.0000	0.1435
Jul 30	2.2976	2.0734	2.2976	0.0000	0.2242
Jul 31	2.3906	2.0678	2.3906	0.0000	0.3229
Aug 1	2.4905	2.0621	2.4905	0.0000	0.4284
Aug 2	2.5802	2.0563	2.5802	0.0000	0.5240
Aug 3	2.6408	2.0503	2.6408	0.0000	0.5905
Aug 4	2.6576	2.0443	2.6576	0.0000	0.6133

Dust (g cm⁻³)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	1.6340	1.6340	1.6340	0.0000	0.0000
Jul 23	1.6015	1.6015	1.6015	0.0000	0.0000
Jul 24	1.5692	1.5692	1.5692	0.0000	0.0000
Jul 25	1.5372	1.5372	1.5372	0.0000	0.0000
Jul 26	1.5345	1.5345	1.5345	0.0000	0.0000
Jul 27	1.5363	1.5363	1.5363	0.0000	0.0000
Jul 28	1.5396	1.5396	1.5396	0.0000	0.0000
Jul 29	1.5449	1.5449	1.5449	0.0000	0.0000
Jul 30	1.5523	1.5523	1.5523	0.0000	0.0000
Jul 31	1.5615	1.5615	1.5615	0.0000	0.0000
Aug 1	1.5714	1.5714	1.5714	0.0000	0.0000
Aug 2	1.5966	1.5966	1.5966	0.0000	0.0000
Aug 3	1.6347	1.6347	1.6347	0.0000	0.0000
Aug 4	1.6685	1.6685	1.6685	0.0000	0.0000

Site: Artemis III Nobile Rim candidate

Solar activity

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	13.48	12.36	18.35	4.87	5.99
Jul 23	13.24	12.36	18.35	5.11	5.99
Jul 24	13.17	12.36	18.35	5.18	5.99
Jul 25	13.52	12.36	18.35	4.83	5.99
Jul 26	13.19	12.36	18.35	5.16	5.99
Jul 27	13.39	12.36	18.35	4.96	5.99
Jul 28	13.46	12.30	18.35	4.89	6.05
Jul 29	13.31	12.28	18.35	5.04	6.07
Jul 30	13.19	12.28	18.35	5.16	6.07
Jul 31	13.35	12.28	18.35	5.00	6.07
Aug 1	13.10	12.28	18.35	5.25	6.07
Aug 2	13.39	12.39	18.35	4.96	5.96
Aug 3	13.50	12.79	18.35	4.85	5.56
Aug 4	13.00	12.28	18.35	5.35	6.07

Radiation (mSv)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	0.0694	0.0413	0.0744	0.0050	0.0331
Jul 23	0.0696	0.0413	0.0744	0.0048	0.0331
Jul 24	0.0694	0.0413	0.0744	0.0050	0.0331
Jul 25	0.0695	0.0413	0.0744	0.0049	0.0331
Jul 26	0.0696	0.0413	0.0744	0.0048	0.0331
Jul 27	0.0695	0.0413	0.0744	0.0049	0.0331
Jul 28	0.0694	0.0400	0.0730	0.0037	0.0331
Jul 29	0.0694	0.0392	0.0728	0.0034	0.0337
Jul 30	0.0694	0.0392	0.0728	0.0034	0.0337
Jul 31	0.0695	0.0392	0.0728	0.0033	0.0337
Aug 1	0.0695	0.0392	0.0728	0.0034	0.0337
Aug 2	0.0694	0.0419	0.0749	0.0055	0.0331
Aug 3	0.0752	0.0472	0.0803	0.0051	0.0331
Aug 4	0.0695	0.0392	0.0728	0.0033	0.0337

Temperature (°C)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	-161.0	-161.0	-161.0	0.0	0.0
Jul 23	-155.2	-155.2	-155.2	0.0	0.0
Jul 24	-155.4	-155.4	-155.4	0.0	0.0
Jul 25	-141.8	-141.8	-141.8	0.0	0.0
Jul 26	-118.1	-118.1	-118.1	0.0	0.0
Jul 27	-106.5	-106.5	-106.5	0.0	0.0
Jul 28	-102.7	-102.7	-102.7	0.0	0.0
Jul 29	-105.9	-105.9	-105.9	0.0	0.0
Jul 30	-106.4	-106.4	-106.4	0.0	0.0
Jul 31	-104.0	-104.0	-104.0	0.0	0.0
Aug 1	-101.9	-101.9	-101.9	0.0	0.0
Aug 2	-104.4	-104.4	-104.4	0.0	0.0
Aug 3	-107.8	-107.8	-107.8	0.0	0.0
Aug 4	-113.5	-113.5	-113.5	0.0	0.0

Moonquakes (per day)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	47.64	47.64	47.29	0.35	0.35
Jul 23	46.46	46.46	46.58	0.12	0.12
Jul 24	45.29	45.29	45.87	0.58	0.58
Jul 25	44.12	44.12	45.16	1.04	1.04
Jul 26	43.28	43.28	44.65	1.37	1.37
Jul 27	43.28	43.28	44.65	1.37	1.37
Jul 28	43.28	43.28	44.65	1.37	1.37

Jul 29	43.28	43.28	44.65	1.37	1.37
Jul 30	43.28	43.28	44.65	1.37	1.37
Jul 31	43.71	43.71	44.91	1.20	1.20
Aug 1	44.88	44.88	45.62	0.74	0.74
Aug 2	46.06	46.06	46.33	0.28	0.28
Aug 3	47.23	47.23	47.04	0.19	0.19
Aug 4	48.40	48.40	47.76	0.65	0.65

Micrometeorite flux ($\times 10^{-15}$)

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	2.0425	2.0418	2.0425	0.0000	0.0006
Jul 23	2.0391	2.0374	2.0391	0.0000	0.0017
Jul 24	2.0372	2.0328	2.0372	0.0000	0.0044
Jul 25	2.0384	2.0281	2.0384	0.0000	0.0103
Jul 26	2.0455	2.0233	2.0455	0.0000	0.0223
Jul 27	2.0628	2.0183	2.0628	0.0000	0.0444
Jul 28	2.0950	2.0133	2.0950	0.0000	0.0817
Jul 29	2.1467	2.0081	2.1467	0.0000	0.1386
Jul 30	2.2194	2.0028	2.2194	0.0000	0.2166
Jul 31	2.3093	1.9974	2.3093	0.0000	0.3119
Aug 1	2.4058	1.9919	2.4058	0.0000	0.4139
Aug 2	2.4924	1.9863	2.4924	0.0000	0.5061
Aug 3	2.5510	1.9806	2.5510	0.0000	0.5704
Aug 4	2.5672	1.9748	2.5672	0.0000	0.5924

Dust (g cm^{-3})

Date	LIPAS	Flagship	Observed / label	LIPAS Error	Flagship Error
Jul 22	1.5663	1.5663	1.5663	0.0000	0.0000
Jul 23	1.5340	1.5340	1.5340	0.0000	0.0000
Jul 24	1.5019	1.5019	1.5019	0.0000	0.0000
Jul 25	1.4867	1.4867	1.4867	0.0000	0.0000
Jul 26	1.4874	1.4874	1.4874	0.0000	0.0000
Jul 27	1.4891	1.4891	1.4891	0.0000	0.0000
Jul 28	1.4922	1.4922	1.4922	0.0000	0.0000
Jul 29	1.4973	1.4973	1.4973	0.0000	0.0000
Jul 30	1.5044	1.5044	1.5044	0.0000	0.0000
Jul 31	1.5133	1.5133	1.5133	0.0000	0.0000
Aug 1	1.5271	1.5271	1.5271	0.0000	0.0000
Aug 2	1.5675	1.5675	1.5675	0.0000	0.0000
Aug 3	1.6052	1.6052	1.6052	0.0000	0.0000
Aug 4	1.6387	1.6387	1.6387	0.0000	0.0000

- Evaluate LIPAS using the exact same nine criteria to map out the structural and functional differences between the unified system and the separate models.

Phase 3: 14-Day Quantitative Accuracy Study

- Set up the control rules by collecting the target lunar coordinates for the chosen testing sites (equatorial 0°N 0°E, Apollo 11, and Nobile Rim), and note the exact 14-day tracking intervals for Window A (5–18 July 2026) and Window B (planned lunar-day window).
- Extract the true observed / label data from the extended dataset used in the hybrid study (public drivers plus physics and catalog anchors) for those specific locations and dates.
- Generate LIPAS predictions with the locked train2 hybrid backend and the frozen blend weights used for Window A, logging daily forecasts across all six hazards.
- Compile the flagship model benchmarks from the public archives or from the official formulas / Python models as described in Section 4.2.
- Compare LIPAS and flagship outputs to the study labels using absolute error / RMSE as reported in Results §§5.3–5.5.

Phase 4: Professional Mission Readiness Simulations

- Build the operation timelines by mapping out a series of trajectories for a lunar landing or an astronaut EVA path, noting the specific safety limits and duration windows.
- Inject changing conditions by altering the live data streams feeding into the LIPAS dashboard, such as manually turning up hazard levels like spikes in radiation or temperature to simulate rapid environmental changes.
- Monitor the system warnings to see if the interface successfully updates the data metrics, triggers risk flags, and shifts the hazard gauges before the simulated safety limits are breached.
- Evaluate situational awareness by comparing the time and steps needed to track these quick emergency windows on the single LIPAS dashboard against the process of checking multiple separate websites and data systems at the same time.

Phase 5: Hazard Ablation and Sensitivity Analysis

- Establish the test baseline by running a normal predictive sequence in LIPAS with all data streams active, and record the baseline error rates for the forecasts.
- Apply data deficit filters by going into the data ingestion pipeline in the code and blocking a specific real-time data input, such as turning off incoming solar storm data while keeping the radiation model running.
- Measure prediction degradation by running the model with this missing data stream and tracking how the final forecasting results change.

- Quantify cross-hazard impact by recording the increase in prediction errors or changes in error rates. Compare these degraded results against the fully integrated baseline to calculate exactly how much data context from one hazard helps improve the prediction accuracy of another.